

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	1 of 87

IMP/010/011 Code of Practice for Earthing LV Networks and HV Distribution Substations

1. Purpose

The purpose of this document is to ensure the company achieves its requirements with respect to the Electricity Act 1989 (as amended) (the Act), the Electricity Safety, Quality, and Continuity (ESQC) Regulations 2002, the Electricity at Work (EAW) Regulations 1989, the Distribution Licences and The Distribution Code, by laying out the way in which CE will develop efficient, co-ordinated and economical earth systems on the Low Voltage (LV) networks and at High Voltage (HV) distribution substations.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/010/011	1	Code of Practice for Earthing LV Networks
0500	1983	Code of Practice for Earthing

This Code of Practice contains detailed policy and practice on earthing of LV Networks and HV distribution substations. Section 3.20 provides basic information regarding HV and LV earthing of distribution substations.

2. Scope

The Code of Practice applies to all LV networks, services and HV distribution substations in NEDL and YEDL the licensed distributors of CE Electric UK Funding Company Ltd (all referred to hereafter as 'CE'), and to the providers of connections to those networks.

It is not intended for this policy to be applied retrospectively with the exception of locations identify as having inadequate earth networks or those where work is being carried out on the network to satisfy the requirements of the ESQC Regulations 2002 as detailed in section 3.3.2 of this code of practice.

All new networks shall be equipped with earth systems in accordance with this Document

3.0 Policy

3.1 Assessment of Relevant Drivers

The key internal business drivers relating to earthing of networks are:
safety,
financial; and
quality of supply.

The external business drivers relating to the Application of earthing to distribution networks are detailed in the following sections.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	2 of 87

3.1.1 Requirements of the Electricity Act 1989 (as amended)

The Electricity Act 1989 (as amended) ('the Act') lays down the core legislative framework for CE's operations as a distributor. Specifically, it gives force to the ESQC Regulations 2002, and in section 9 creates the key obligation to develop and maintain an efficient, co-ordinated and economical system of electricity distribution. Discharge of this obligation shall be supported in this document by providing guidelines on efficient application of earth systems to the wider network.

3.1.2 Requirements of The Electricity Safety, Quality and Continuity (ESQC) Regulations 2002.

The ESQC Regulations 2002 impose a number of obligations on the business, mainly relating to quality of supply and safety. All the requirements of the ESQC Regulations that are applicable to the application of earthing shall be complied with, specifically:

Reg. No	Text	Application to this policy
3(1)(b)	...distributors...shall ensure that their equipment is so constructed...as to prevent danger...or interruption of supply, so far as is reasonably practicable	This will be achieved by installation of earth systems to standard designs by trained personnel.
6	A...distributor shall be responsible for the application of such protective devices to his network as will, so far as is reasonably practicable, prevent any current, including any leakage to earth, from flowing in any part of his network for such a period that that part of his network can no longer carry that current without danger.	This will be achieved by installation of systems and conductors to provide a low impedance path back to source for earth fault currents as far as is reasonably practicable
7(1)	A generator or distributor shall, in the design, construction, maintenance or operation of his network, take all reasonable precautions to ensure continuity of the supply neutral conductor	This will be achieved by installation of system conductors to minimise damage and accidental disconnection
8(1)	A generator or distributor shall ensure that, so far as is reasonably practicable, his network does not become disconnected from earth in the event of any foreseeable current due to a fault.	This will be achieved by installation of earth systems and conductors of sufficient rating and in such a manner as to prevent disconnection
8(2)a	A generator or distributor shall, in respect of any high voltage network which he owns or operates, ensure that – the network is connected with earth at, or as near as is reasonably practicable to, the source of voltage but where there is more than one source of voltage in that network, the connection with earth need only be made at one such point	This will be achieved by design of the network

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	3 of 87
8(2)b	A generator or distributor shall, in respect of any high voltage network which he owns or operates, ensure that – the earth electrodes are designed, installed and used in such a manner so as to prevent danger occurring in any low voltage network as a result of any fault in the high voltage network	This will be achieved through standard designs and where appropriate the separation between the HV and LV earth electrodes.			
8(3)b	A generator or distributor shall, in respect of any low voltage network which he owns or operates, ensure that - every supply neutral conductor is connected with earth at, or as near as is reasonably practicable to, the source of voltage except that where there is only one point in a network at which consumer's installations are connected to a single source of voltage, that connection may be made at that point, or at another point nearer to the source of voltage	This will be achieved by design of the network			
8(4)	A consumer shall not combine the neutral and protective functions in a single conductor in his consumer's installation	Those who need to provide information to customers made aware of this requirement through this Code of Practice			
9(1)	This regulation applies to distributors' low voltage networks in which the neutral and protective functions are combined	This will be achieved by installation of earth systems to the requirements laid down in this Code of Practice			
9(2)	(2) In addition to the neutral with earth connection required under regulation 8(3)(b) a distributor shall ensure that the supply neutral conductor is connected with earth at - (a) a point no closer to the distributor's source of voltage (as measured along the distributing main) than the junction between that distributing main and the service line which is most remote from the source; and (b) such other points as may be necessary to prevent, so far as is reasonably practicable, the risk of danger arising from the supply neutral conductor becoming open circuit.	This will be achieved by installation of earth systems to the requirements laid down in this Code of Practice			
9(4)	The distributor shall not connect his combined neutral and protective conductor to any metalwork in a caravan or boat	This will be achieved by following the guidance in this Code of Practice			

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	4 of 87
10	This regulation requires that any metalwork enclosing, supporting or otherwise associated with his equipment in a network and which is not intended to serve as a phase conductor is, where necessary to prevent danger, connected with earth (there are exceptions for metalwork connected to wood poles more than 3 metres clearance from ground level)		This will be achieved by following the guidance in this Code of Practice		
24(4)	Unless he can reasonably conclude that it is inappropriate for reasons of safety, a distributor shall, when providing a new connection at low voltage, make available his supply neutral conductor or, if appropriate, the protective conductor of his network for connection to the protective conductor of the consumer's installation		This will be achieved by following the guidance in this Code of Practice. This regulation is expanded on in the following paragraphs		

Part V11 Paragraph 24, Sections (4) & (5) of the ESQC Regulations states that: -

(4) "Unless he can reasonably conclude that it is inappropriate for reasons of safety, a distributor shall, when providing a new connection at low voltage, make available his supply neutral conductor or, if appropriate, the protective conductor of his network for connection to the protective conductor of the consumer's installation."

(5) In this regulation the expression "new connection" means the first electric line, or the replacement of an existing electric line, to one or more consumer's installations

Paragraph 24 of the ESQC Regulations specifically excludes the connection of the distributors combined neutral and protective conductors to any metalwork in a caravan or boat. Further information on locations where an earth terminal should not be offered are given in Section 3.15 of this Code of Practice.

The general CE policy is to ensure that an earth terminal can be provided so that, when an installation has been designed to the current edition of the IEE Regulations for Electrical Installations (BS7671), a customer's fuse will operate in the event of an earth fault on the customer's installation.

In order to ensure adequate current flows, the best method is to use a continuous metallic return path between the customer's earth terminal and the transformer neutral point. The soil is inadequate for the purpose of carrying earth fault current because of its high resistance. Metallic water pipes, although often used in the past, are unsuitable because of the increasing use of non-metallic pipes and inserts, and the potential high resistance to earth.

The continuous metallic return path will be provided by cable sheath, by separate cable conductor, by protective multiple earthing (PME) or by protective neutral bonding (PNB). PME and PNB are means whereby the neutral conductor, apart from its normal function of carrying load current, provides the metallic return path for earth fault currents from the customer's premises to the substation transformer neutral. The adequacy of the return path is determined by its loop impedance.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	5 of 87

In addition the use of protective multiple earths, interconnection of neutrals and good load balancing where possible will normally ensure that, even if the neutral breaks, any customer will be left with at least one earth connection on that portion of the neutral still connected to the installation. This will in most cases tend to hold down the potential of the neutral and consequently the voltage applied to the frames of customer owned metallic apparatus, to a reasonable value.

3.1.3 The Health and Safety at Work etc Act 1974

Section 2(1) states that 'It shall be the duty of every employer to ensure, so far as is reasonably practicable, the health, safety and welfare at work of all his employees'. Section 3(1) also states that 'It shall be the duty of every employer to conduct his undertaking in such a way as to ensure, so far as is reasonably practicable, that persons not in his employment who may be affected thereby are not thereby exposed to risks to their health or safety'.

This is addressed in this code of practice by specifying the requirements of earth systems.

3.1.4 Electricity at Work Regulations

The UK Electricity at Work Regulations (Statutory Instrument 1989, No 635) was made under the 1974 Health and Safety at Work Act. Regulation 8 relates to earthing which quotes "Precautions shall be taken, either by earthing or by other suitable means, to prevent danger arising when any conductor (other than a circuit conductor) which may reasonably foreseeable become charged as a result of either the use of a system, or a fault in a system, becomes so charged; and, for the purposes of ensuring compliance with this regulation, a conductor shall be regarded as earthed when it is connected to the general mass of earth by conductors of sufficient strength and current-carrying capability to discharge electrical energy to earth."

These requirements are met within this code of practice by common referencing to earth, connection of plant to earth and equipotential bonding.

3.2 Key Requirements

CE policy is that all networks shall be so constructed such that an earthing terminal can be made available at every LV supply point and that this will provide a solid metallic return from the earth terminal of the installation back to the substation transformer neutral terminal. Earthing on all LV networks shall be carried out in accordance with the policy and procedures detailed in this Code of Practice. These procedures all comply fully with the ESQC Regulations (2002), and it will not be necessary for field staff to make a detailed study of the regulations.

The details for implementing this policy are set out in the appropriate sections of this Code of Practice. Matters relating to contact with customers are to be found in Appendix A and B of this Code of Practice.

3.2.1 New Networks

All new CE underground and overhead LV networks shall be constructed to the earthing standard required by this Code of Practice so that normally all new customers can be

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	6 of 87

provided with an earthing terminal. (In a few special circumstances it may not be appropriate to offer the terminal examples are discussed in section 3.15 of this document)

3.2.2 Existing Networks

The implementation of this policy ensures that CE LV customer can have a solid metallic earth-fault return path provided by one of the following means:

- (a) Protective multiple earthing (PME)
- (b) The electrically continuous sheaths of separate neutral and earth (SNE) cables
- (c) Protective neutral bonding (PNB)

The choice between these alternatives will depend on the nature of the existing distribution system and the cost of providing the earthing terminal.

(In a few special circumstances it may not be appropriate to offer the terminal see sections 3.15 for details)

3.3 Implementation

3.3.1 New Networks

New underground and overhead networks shall be constructed in accordance with the requirements of this Code of Practice. All customers on such networks will be offered the use of a CE earth terminal connected to the neutral conductor.

New underground networks shall normally use combined neutral and earth (CNE) cables. However, it shall be permissible to use separate neutral and earth (SNE) cables for services or networks where this is beneficial. In particular, for customers whose installations may require special attention such as those identified in 3.15 of this Code of Practice, an economic evaluation should be carried out of the alternative methods of earthing. For customers in this category, and if the network is suitable, we may offer an SNE supply.

New overhead networks shall use aerial bundled conductors in accordance with the code of practice NSP/004/041 LV ABC Networks.

PNB may be adopted for a single customer supplied via an underground or overhead service from a substation. However it is acceptable to provide PME to a single customer and this method should be adopted where practical.

3.3.2 Existing Networks

The earthing arrangements on LV networks should be checked to see whether it is to the standard described in this Code of Practice whenever any of the following situations occur:

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- (a) Connecting a new substation into the network, or recovering a substation from the network.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	7 of 87

- (b) Changing a transformer in an existing substation.
- (c) Reinforcing or extending the network.
- (d) Repairing the network.
- (e) Customers' earthing facilities which are provided via metallic water pipes are about to be rendered ineffective by the insertion of non-metallic pipes in distributors or service pipes by the Water Company or plumbing contractors.
- (f) When the theft of aerial earth-wires has rendered earthing ineffective.

After review, action shall be taken as follows:

- (i) If the network is already covered by PME or PNB, or comprises underground cables with electrically continuous separate sheaths or overhead lines with aerial earth-wires throughout, the networks shall require no modification except for the installation of additional earth-points and compression jointing of connections on neutrals and aerial earth-wires to reduce the risk arising from conductor breakage or theft.
- (ii) If the network comprises cables having electrically discontinuous sheaths but the number of discontinuities is small, the sheaths shall be made continuous by bonding across the discontinuities.
- (iii) In all other cases PME shall be applied to the network in accordance with the requirements of this Code of Practice.

3.4 Provision of CE Earth Terminals

3.4.1 Provision of CE earth terminals on networks with electrically continuous sheaths or separate earth conductors

Where CE has not already provided earth terminals for customers on networks with electrically continuous sheaths or separate earth conductors, we shall provide an earth terminal.

Such opportunities may arise when:

- (a) We provide a new service.
- (b) The customer or their electrical contractor asks us to provide an earth terminal.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	8 of 87

(c) We change or move the service intake-unit.

(d) We review the earthing arrangements on the network as detailed in Section 3.3.2.

3.4.2 Provision of CE earth terminals on networks to which PME has been applied

Customers on networks or parts of networks to which PME has been applied shall be offered the use of a CE earth terminal. Before such a terminal can be used, however, customers must ensure the bonding within their installations complies with the requirements laid down in Appendix A.

3.4.3 Adequacy of Existing Bonding

Where a new PME terminal is to be provided the customers bonding must comply with the requirements of Appendix A and the current edition of BS7671 (IEE Wiring Regulations)

For customers with existing PME terminals, their bonding should be in accordance with the PME requirements at the time the terminal was made available. In general, this bonding will be 10mm² however; some installations may have 6mm² bonding. This 6mm² bonding will be acceptable where the terminal is in use.

When an existing terminal is in use prior to a cut-out change or when an inspection is carried out on an existing installation with 6mm² bonding, we will issue a safety notice to the customer to recommend that they change the bonding from 6mm² to 10mm². PME installations with 2.5 mm² or 4mm² bonding will not be acceptable, and measures must be taken to ensure that the customer's installation is brought up to the requirements of Appendix A.

3.4.4 Provision of an earth terminal after a cut-out change

Where no other work is involved other than a cut-out is changed for whatever reason the earthing arrangements shall remain unchanged so that if the existing cut-out provided an SNE earth facility the replacement cut-out shall provide an SNE earth facility even if it can be confirmed that the SNE network has been converted to PME unless the customer can confirm in writing that their installation meets the requirements of appendix A.

3.4.5 Short-term safeguards for customers on substantial existing networks without adequate earth- return facilities

There may be networks, such as those purchased from other suppliers, which do not have adequate earthing facilities as defined in Section 3.2.2. In the interim period before these networks are suitably converted, problems may arise concerning the earthing of existing customers' installations. These problems will fall into four groups and will be dealt with as follows:-

- (i) On adoption or identification of networks with inadequate earth facilities the customers affected shall be advised in writing to install a residual current device. Asset Management will be informed and initiate a work programme to update the network such that earthing facilities defined in Section 3.2.2 can be provided to the customers.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	9 of 87

- (ii) A request from a customer or his contractor for an earthing terminal. Appendix B note [2] bullet point 1, describes the procedure to be followed if the network is not suitable to enable us to offer an earth terminal.
- (iii) CE staff may visit customers' installations for miscellaneous reasons such as the reconnection of a supply. If the installation is found to be inadequate then the procedure in Appendix B should be followed to notify the customer.
- (iv) If a contractor carries out alterations to a customer's installation and the earthing of that installation is found to be inadequate then the new installation shall not be connected until the earthing arrangement conforms to the current edition of the IEE Regulations for Electrical Installations (BS7671) and that situation is confirmed by issue of the relevant Part 7 BS7671 certificate or by other means of written confirmation that may be agreed.

3.5 PME and Multiple-Earthing Approvals

In the past it was necessary to apply to the Department of Energy for approval to apply MEN (Multiple Earthing of the Neutral) or PME to networks. This has not been the case for many years. Consent was also required to apply PME to private networks.

The Electricity Safety, Quality and Continuity (ESQC) Regulations 2002 allows all Distributors (not necessarily Licensed Distributors) to apply PME to their networks. However it is still necessary for such non Regional Electricity Company networks to meet the requirements of the ESQC regulations.

3.6 Maximum Earth Values at Secondary Distribution Substations

Where HV and LV earth electrodes are segregated the following maximum values shown in paragraph 3.20.1.2 (table 3.20.1.2) shall apply.

See Section 3.20 for more detailed information regarding earthing at ground and pole mounted substations including standard arrangements for ground mounted substations

3.7 Earthing Arrangements on New LV Networks

3.7.1 New Underground Networks

New underground networks shall normally use CNE cables with copper neutral/earth conductors and PME will be adopted. However, it shall be permissible to use SNE cables for services or networks where this is beneficial for safety or economic reasons. In particular, this could be for customers whose installations may require special attention such as those identified in Section 3.15 of this Code of Practice.

The earthing arrangements at new substation must comply with Section 3.20.

For PME networks, obligatory connections of the neutral conductor to earth electrodes or neutral conductors of other distributors shall be made at the final insulated end of every branch of the distributor. A multi-service line supplying up to four adjacent customers need not be considered as a branch of the distributor for this purpose. Where a service supplying up to four adjacent customers is 40 metres or more in length an earth electrode must be installed and connected to the neutral at or near the last service termination.

If the HV and LV earth electrodes have been separated at the substation the overall resistance to earth of the HV earth electrode and that of the neutral conductor shall not

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	10 of 87

exceed 20 ohms. Measurements of resistance will be made by an approved method. If the resistance of the HV earth electrode to earth exceeds 20 ohms then additional earth electrodes shall be connected to reduce the resistance. If the resistance of the neutral conductor to earth exceeds 20 ohms then additional earth electrodes shall be connected at convenient points on the network to reduce the resistance. The method of installation of earth electrodes shall be as described in 3.9.

If the HV and LV earth electrodes have been combined at the substation, because they meet the conditions in Section 3.20.5 no measurement of neutral-to-earth resistance need be made.

Drawing C952986 shows typical arrangements of LV network earthing.

3.7.2 New Overhead Networks

New overhead networks shall normally use CNE Aerial Bundled Conductor (ABC) in accordance with the code of practice NSP/004/041 LV ABC Networks and PME will be adopted.

It shall be permissible to use separate neutral and earth conductors for services or networks where this is beneficial, however the practice of making SNE overhead customers dependant on CNE overhead circuits shall be avoided due to possible inadequacy of customers internal bonding. In particular, SNE could be used for customers whose installations may require special attention such as those identified in Section 3.15 of this Code of Practice.

The earthing arrangements at new substation must comply with Section 3.20

For PME networks, the neutral conductor shall be connected to earth electrodes or to the neutral conductors of other distributors at the final supports of every branch of the distributor. Additional earth electrodes shall be installed at the end of each section of ABC distributor, at junctions between cable and overhead lines and at all service distribution boxes. Additional earths shall be installed at intervals of not more than eight span-lengths along open wire distributors. A multi-service line or undereaves wiring supplying up to four adjacent customers need not normally be considered as a branch of the distributor for this purpose. However, where a single service exceeds three overhead spans in length an additional earth electrode shall be installed either on the overhead service termination, where the connection is made to the undereaves wiring installation or at the final support of the overhead line.

If the HV and LV earth electrodes have been separated at the substation the resistance to earth of the HV earth electrode shall not exceed 20 ohms and the overall resistance to earth of the neutral conductor shall not exceed 20 ohms. Measurements of resistance will be made as indicated in Appendix D. If in the first instance the resistance of the HV electrode to earth exceeds 20 ohms then additional earth electrodes shall be connected to reduce the resistance. If the resistance of the neutral conductor to earth exceeds 20 ohms then additional earth electrodes shall be connected at convenient points in the network to reduce the resistance. The method of installation of the earth electrodes shall be as described in Section 3.9.

If there is any overhead line in the HV circuit between the ground mounted HV substation and the source substation the HV electrode and the LV earth electrode shall be separated unless the conditions in section 3.20.5 are met. The HV earth electrode and LV earth electrodes shall not be combined at pole mounted transformers without written permission from the Policy Manager.

Good load balance along an LV feeder will assist in containing neutral voltage rise in the event of an open circuit neutral conductor.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	11 of 87

Drawing C952986 shows typical arrangements of LV network earthing.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	12 of 87

3.8 Earthing Arrangements on Existing LV Networks

3.8.1 Underground Networks with Electrically Continuous Sheaths

Existing earthing arrangements of such networks are satisfactory and will not require modification.

Any extensions or modifications including repairs to these networks will employ either CNE or SNE cables, when SNE cable which has an insulated outer sheath is used an earth electrode shall be installed at each service joint position.

When repairs or alterations are carried out on SNE networks, SNE cable may be used in the following circumstances:

- (a) Where it is unlikely that customers will be persuaded to bond their installation's to conform to the PME requirements as detailed in Appendix A.
- (b) Where we expect to have difficulty in gaining access to customers' premises to see that the bonding conforms to the requirements of Appendix A.
- (c) Where a service alteration to an SNE service is required

The following arrangements may occur in the network:

(i) CNE Cable Supplied via an SNE Cable

The neutral conductor of the CNE cable shall be connected to the neutral conductor and sheath or earth conductor of the SNE cable at the transition joint. Further earth electrodes shall be installed on the CNE cable section in accordance with Section 3.7.1.

Drawing C952994 Sheet 1 and C952997 Sheet 2 show typical arrangements.

The work completed above will result in the CNE cable section and that part of the SNE cable lying between the substation and the CNE cable having been converted to PME. But neither the branches of this SNE cable, nor any other distributors on the network will have been converted unless the neutral conductor has been earthed at an appropriate position on the branch or distributor.

Drawing C953019 shows typical applications of CNE cable

(ii) SNE Cable Supplied via a CNE Cable

The neutral conductor of the CNE cable shall be connected to earth electrodes or the neutral conductors of other distributors at the end of every branch. At the transition joint it will be connected to the neutral conductor and metallic sheath of the SNE cable.

Existing SNE customers' may retain an SNE earth where the following conditions are met:

The circuit is interconnected with another substation via SNE cable.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	13 of 87

The SNE cable is 50 metres or more in length, whose metallic sheath is in close contact with earth, as it will normally provide an adequate connection with earth (10 ohm or less) to limit the rise of potential under open circuit neutral.

Where the SNE cable is less than 50 metres in length and not interconnected with another substation then an earth electrode connected to the cable sheath shall be installed at each service joint position and at the end of each branch or the cable shall be converted to PME and the bonding checked at existing properties. Further earth electrodes need be connected only to ensure that the overall resistance to earth of the neutral of the LV network complies with Section 3.7.1.

Drawing C953011 Sheet 1 and C953015 Sheet 2 show typical arrangements.

The work completed above will result in the CNE cable section having been converted to PME. The SNE cable section will not have been converted to PME unless the neutral conductor has been earthed at additional positions.

Drawing C953019 shows typical applications of CNE cable

(iii) Section of CNE Cable Inserted between Two Sections of SNE Cable

The neutral conductor of the CNE shall be connected to the neutral conductor and metallic sheath or earth conductor of the SNE cables at the transition joints.

Existing SNE customers' on the SNE cable most remote from the substation may retain an SNE earth where the following conditions are met:

The circuit is interconnected with another substation via SNE cable.

The SNE cable is 50 metres or more in length, whose metallic sheath is in close contact with earth, as it will normally provide an adequate connection with earth (10 ohm or less) to limit the rise of potential under open circuit neutral.

Where the SNE cable is less than 50 metres in length and not interconnected with another substation then an earth electrode connected to the cable sheath shall be installed at each service joint position and at the end of each branch or the cable shall be converted to PME and the bonding checked at existing properties. Further earth electrodes need be connected only to ensure that the overall resistance to earth of the neutral of the LV network complies with Section 3.7.1.

The work completed above will result in the CNE cable section and that part of the SNE cable section lying between the substation and CNE cable section having been converted to PME. Any branches of the SNE cable nearer the substation and all the SNE cable more remote from the substation will not have been converted to PME unless additional neutral to earth connections have been made at appropriate positions.

Drawing C953019 shows typical applications of CNE cable.

(iv) Cable Overlay/Replacement in SNE Cable Network

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	14 of 87

Where it is necessary to overlay a section of SNE cable in an SNE network 3 core CNE waveform cable will be used unless it can be shown to be more economic to use 4 core SNE cable in which case 4 core SNE cable may be used. Where the SNE cable which is replaced had a metallic sheath which is in close contact with earth the disconnected cable will be retained and the metallic sheath used as an additional earth electrode to help maintain the network earth. The disconnected cable must be marked up on the cable records as an earth electrode.

Where 3 core CNE waveform or 4 core SNE waveform cable is used the following additional procedure will be adopted:-

At suitable service joint positions (Approx. every 50m) cross-bond the copper earth screen to the metallic sheath of the disconnected cable with 35mm sq cu insulated conductor using a substantial sweated connection or an earth clamp which complies with the requirements of BS951. This will help reduce the overall network earth impedance.

If 4 core SNE waveform cable is used the following procedure will be adopted:-

The tail end of the 4 core waveform cable will be cap-ended with an earth electrode to PME standard. Both SNE and CNE earth facilities can continue to be made available along the route in accordance with Section 3.8.1 (v).

(v) Provision of CE Earth Terminals on mixed CNE and SNE Cable Networks

CE earth terminals, connected to the neutral conductor, will normally be provided to customers connected to CNE cable sections unless they are precluded under the ESCQ regulations (boats and caravans) or are a "Special Case" as defined in Section 3.15. The customers' installations must comply with the bonding and earthing conditions of Appendix A before connection to the earthing terminal may be made.

Existing services on the SNE cable section may either be provided with, or may retain, CE earth terminals connected to the sheath or earth conductor. When PME has been applied to part of the network, all new services on the SNE sections of the network will be provided by CNE service cables and the customers offered the use of a CE earth terminal connected to the neutral; their installations must then comply with the PME bonding conditions of Appendix A. If the metallic sheath of the SNE cable section is in close contact with earth it will be necessary to bond the neutral conductor of the CNE service cable to the neutral and metallic sheath of the SNE cable section only at the service joint. In cases where the SNE sheath is insulated from earth or is less than 9 metres in length then an earth electrode must be installed and connected to the neutral at the service joint, unless a similar connection has previously been made on the branch concerned at a point farther from the supply transformer. Note -The 9 metres of SNE cable is equivalent to an earth rod which would need to be installed at the most remote point on the network or at the most remote service position to enable PME supplies to be provided from the network.

If CNE cables are used to replace existing lengths of SNE cables and existing SNE service cables are transferred to the CNE cable, any existing CE earth terminals must be replaced by earth terminals connected to the neutral conductor. The metallic sheath of the SNE

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	15 of 87

service cable shall then be bonded to the neutral conductor of the CNE cable at the service joint. Customers' installations must then comply with the requirements of Appendix A.

3.8.2 Underground Networks with Electrically Discontinuous Sheaths

Existing earthing arrangements of such networks are unsatisfactory and shall be brought to the required standard by a programme.

When it is decided to apply PME to the network the earthing arrangements at the substation will be reviewed made to comply with Section 3.20 where practical

The neutral conductor of the distributor shall be connected to earth electrodes, or to the neutral conductors of other distributors, at the ends of all branches. Alternatively, where it is more convenient, earth connections may be made at points not nearer to the substation than the remote service at which a CE earth terminal is to be provided. The connection may be made either at the most remote PME joint position or at the corresponding service intake.

Where the earth connection is made at the customers service intake an earth electrode shall be installed.

The customer's earthed metalwork or gas and water service pipes shall not be used for this purpose.

3.8.3 Overhead Networks with Aerial Earth Wires

The earthing arrangements of such networks are satisfactory in the short term. When, however, the network is reviewed in accordance with Section 3.3.2 additional earth electrodes connected to the earth wire and neutral conductor shall be installed at those positions at which earth electrodes would have been required if the network were being converted to PME and all neutral connections made with compression joints, duplicate split bolt connectors mechanical connectors or ABC insulation piercing connectors.

The work completed above will result in the overhead section having been converted to PME future services should be offered as PME whilst existing SNE services may be retained.

Note- Single bolt insulation piercing connectors are suitable for service neutral connections but distributor neutral connections must utilise double bolt insulation piercing connectors.

3.8.4 Overhead Networks without Aerial Earth Wires and which have not been converted to

PME or PNB

Existing earthing arrangements of such networks are unsatisfactory and shall be brought to the required standard in accordance with an agreed programme or when any of the opportunities in Section 3.3.2 arise.

PME shall be applied to the networks and the earthing arrangements at the substation supplying the network shall be made to comply with the requirements of Section 3.20 where practical.

The neutral conductor of the network shall be connected to an earth electrode, or to the neutral conductor of another distributor, at the end of every distributor or branch to which PME is to be applied. These connections shall normally be made at the final support of the distributor in order that all customers may be offered a CE earth terminal.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	16 of 87

Group services or under-eaves wiring supplying up to 4 adjacent customers need not be considered as branches of the distributor for this purpose.

Additional earth electrodes shall be installed at intervals of not more than eight span lengths along the route of a long overhead distributor. All neutral connections shall be made using compression joints or duplicate mechanical connectors.

Where LV underground cables are connected in overhead lines the metallic casing of cable boxes and any metallic cable sheaths shall be connected to the neutral conductor at both ends of the underground cable sections. Line tap type neutral connections on tier type cable boxes shall be replaced with compression lugs fitted with double locking nuts.

Where an LV isolator is installed in the run of an overhead line the neutral blade of the isolator shall be removed and replaced by a bolted link or compressed jumper connection.

If the HV and LV earth electrodes have been separated at the substation, the overall resistance to earth of the neutral conductor must not exceed 20 ohms. The measurement of resistance shall be made by an approved method and additional earth electrodes installed at convenient points on the network as necessary to achieve the required value.

If the HV and LV earth electrodes have been combined at the substation no further measurements of neutral to earth resistance need be made.

3.8.5 Overhead and Underground Networks which have been converted to PME

Existing earthing arrangements of such networks are satisfactory and require no modification. Any extensions to these networks shall be made in accordance with the ESQC Regulations (2002), irrespective of the form of approval under which they were initially constructed. Existing CE earth terminals may be retained but new ones will be provided only if a customer's installation complies with the conditions of Appendix A.

3.9 Earth Electrodes on PME Systems

All earthing materials shall be in accordance with CE Electric material specification NPS/002/001 Specification for Earthing Materials.

When PME is applied to an LV network obligatory earth electrodes must be connected to the neutral conductor of a distributor at points remote from the substation. As a minimum, these electrodes shall consist of a driven rod, 1.2m or strip electrode of similar area. The driven rod electrode is preferred and plate electrodes should no longer be used. Where it is not practical to install a driven rod because of the proximity of other services or cables in the footpath the preferred alternative is install a 2 metre length of bare 70mm² (32mm² for service joints) copper conductor around the base of the joint hole or trench as illustrated in figure 3.9 below. Alternatively if, at the position where the neutral is to be connected to an earth electrode, it is connected to the metallic sheath of a cable or the metallic case of a link box or pillar which is in contact with earth then a separate earth electrode need not be installed providing the overall resistance to earth is less than 20 ohms. The sheaths and armouring of hessian served cables may be assumed to be in contact with earth. The earth resistance of individual earth electrodes need not be determined.

Where earth electrodes are installed at substations they shall consist of driven rods or stranded copper conductor or copper strip, the driven rod is preferred. The earth resistance of these electrodes will normally have to be measured at pole-mounted substations and, in combination with HV cable sheaths etc, at ground mounted substations.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	17 of 87

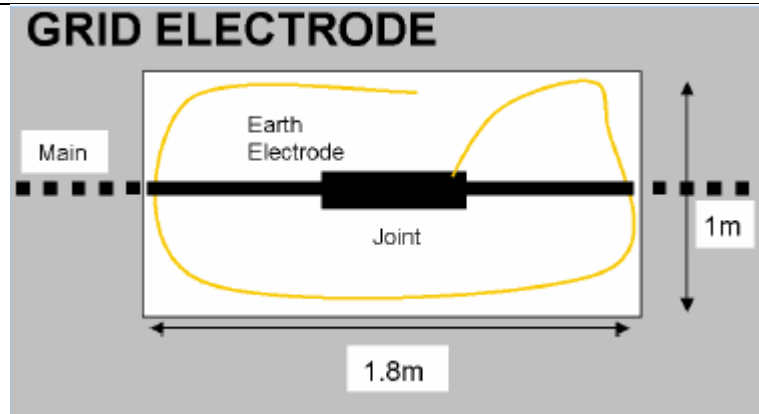


Figure 3.9 earth electrode used as an alternative to an earth rod

Connections to earth electrodes shall be insulated from earth to a depth of at least 600mm below ground level using the conductors specified in Appendix E. Where 9m separation of earth electrodes is necessary, then the connection cable in the ground should be insulated irrespective of its depth.

Where additional earth electrodes have to be installed, either at the substation or on the LV network, to achieve a required value of resistance it will generally be more effective to use driven rods than plate or strip electrodes.

Where additional electrodes and the main earth electrode are connected together, they should be separated from each other and from any cables by a distance not less than the depth to which they are driven, and should be connected together using copper tape or bare copper cable of the same cross sectional area as the normally insulated earth electrode conductor. The additional electrodes should be placed so that any necessary separation of the earth electrode systems is maintained.

3.10 Arrangements at Link Boxes and Pillars on PME Networks

Where metal link boxes are installed on CNE cables the metal case of the box must be connected to the neutral conductor of the LV network. New metal link boxes for use on CNE cable networks are provided with suitable facilities for the neutral-to-case bond. If metal link boxes are installed so that the cases are not in close contact with earth or when plastic link boxes are installed then an earth electrode must also be installed and connected to the case.

On PME networks using SNE cables the metallic case of the link box or pillar will generally be connected to the cable sheaths. Provided that the metallic sheath of at least one cable is electrically continuous and is connected to the neutral conductor of the network at a convenient position, this connection is adequate. If the cable sheaths are not electrically continuous or not connected to the case of the link box or pillar, suitable connections must be provided either by bonding the neutral conductor to the case within the link box or pillar, or by connecting the sheath to the link box or pillar and to the neutral conductor at a convenient position (e.g. a joint or service termination). If the metallic case of a link box or pillar is not in contact with earth, either directly or by connection to a cable sheath which is earthed, then an earth electrode must be installed and connected to it.

All neutral links should be fitted permanently in position by bolting even where the phase links are removed for normal operation.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	18 of 87

All metallic link box pits should also be bonded to the metal case of the link box and the neutral conductor.

Where non-metallic link boxes are used in conjunction with metallic link box pits, there shall be a bond between the metallic link box pit and the neutral conductor.

3.11 Street Lighting and Other Street Furniture

3.11.1 The Application of PME

The following paragraphs provide general guidance on the provision of earth electrodes for street lighting supplies Code of Practice DSS\01\0016 Guidance on Engineering Recommendation G39 (Earthing Arrangements for Street Lighting Furniture) should be consulted for more detailed information.

The application of PME to street furniture was previously governed by an Exemption to the Electricity Supply Regulations. These have now been superseded by the 2002 ESQC regulations which do not differentiate licensed & non licensed operators, but make reference to Street Electrical Fixtures with an expectation to comply with the current edition of "ER G12 Application of PME to LV networks".

The exemption permits the use of PME on street lighting authority owned CNE cables and permits bonding with 6mm² copper equivalent conductor rather than 10 mm².

Under the exemption it is not permissible to omit the installation of electrodes on street lighting/ furniture distributors of 40m or less length. Consequently, if more than one item of street furniture is along the route, the neutral / earth electrode must be installed at the source of the last service, or preferably at the last service position. Single services to street furniture from a PME distributor do not need an electrode, provided that the distributor neutral is earthed either at or beyond the service joint position.

It is not permissible to consider metallic street furniture to be earth electrodes, although they will carry neutral current if earthed and within a PME system.

The exemption to the regulations applies to street furniture supplied by DNOs or street lighting authorities when using CNE cable. C953029 shows a street lighting authority CNE distributor fed from a PME service to a street lighting pillar. At the street lighting pillar, the DNO's PME neutral will be connected to the customers' earthing terminal, to the pillar and associated metalwork. Each column will be bonded to the neutral of the street lighting cable. At the last column an earth electrode will be installed.

Private installations are not covered by the exemption. These and local authority installation using SNE cables which are supplied from PME services must be SNE. A typical installation is shown in C953031.

3.11.2 Street Lighting Brackets mounted on Poles or Buildings.

The earth terminal and any external metalwork of street lighting fittings mounted on buildings or wooden poles will be connected to the PME earth terminal. The only exception is if the fitting is double insulated. Bond size will be 6mm².

3.11.3 Roadside and Other Housings Accessible to the Public.

Examples are public telephones, pedestrian crossing bollards, and ticket machines. Equipment of this type should be Class II (double insulated) or equivalent construction. No

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	19 of 87

mains derived earthing terminal is required, neither is a residual current device needed for earth fault protection.

3.11.4 Existing Separate Street-Lighting Mains with Inadequate Earth-Return Facilities

The earthing arrangements on these distributors shall be brought to the standard required by the Electricity Safety, Quality and Continuity Regulations 2002. This shall be done either at the request of the Local Authority or as the result of the earthing improvements on the electricity distribution network.

The sheaths of any underground street lighting cables and the metalwork of any link boxes, pillars or pole boxes will be connected to the neutral conductor.

3.11.5 Street-Lighting Services from Electricity Networks with Inadequate Earth Return Facilities

Where individual street-lighting services are provided from distributing mains with inadequate earth-return facilities two courses of action are possible: -

- (i) Bring the earthing arrangements of the whole network up to the standard required by this Code of Practice.
- (ii) Apply the Electricity Safety, Quality and Continuity Regulations 2002 to those parts of the network from which street-lighting services are fed.

Generally the former course of action will be preferable as the benefit of improved earthing facilities may be offered to all customers.

In some cases, however, the latter course of action may be necessary. On every branch or section of a distributing main to which the relevant street furniture is connected the neutral conductor shall be bonded to the non-current carrying metalwork of the relevant equipment with the exception that metal doors of concrete and plastic columns shall not be so bonded where the supply to the column is derived from a PME network. The neutral conductor of the distributor will also be connected to earth or to the neutral of another distributor at a point farther from the supply transformer than the most remote street lighting column etc, at which point an earth terminal is to be provided.

In any case where looped services are provided between street lamps an earth electrode shall be installed and connected to the neutral conductor at the final lamp position. It is not permissible to consider metallic street furniture to be earth electrodes.

3.11.6 Street Lighting Columns with Inaccessible Cable Sheaths

The bonding of the non-current carrying metalwork of street lighting columns includes the bonding of the supply cable armouring and metallic sheath to the neutral conductor. Cases will arise where the street lamp service cable will be terminated below the column door opening and therefore inaccessible for the fitting of a bonding clamp. In such cases the armouring and metallic sheath of the service cable need not be bonded to the neutral conductor.

This exception is permitted because the possibility of being able to touch the cable sheath is extremely remote and for it to be at a potential there would have to be an open-circuit neutral conductor fault in existence at the time.

3.12 Supplies to Customers Direct from Substation

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	20 of 87

3.12.1 Ground-Mounted Substations

If the customer's earth-continuity conductors (including any cable sheaths employed as such) are marshalled at a point in or near the substation a final earth-continuity conductor shall be provided to connect these conductors to the transformer neutral either directly or via the general substation bonding and earthing conductors.

This conductor shall have a copper equivalent cross-sectional area at least half that of the largest outgoing phase conductor, subject to a minimum value of 16 mm². If the cable(s) supplying the customer have electrically continuous metallic sheaths these may be used for the final continuity conductor. The earthing arrangements at the substation shall comply with conditions of Sections 3.20.5 and appendix E.

Where the customer's supply terminals are remote from the substation and a CNE cable has been used to provide the supply it will be inconvenient to install a separate earth-continuity conductor and in such cases PME shall be adopted. The earthing arrangements at the substation shall comply with the conditions of Section 3.20.5. At the customer's supply terminals the neutral conductor shall be connected to an earth electrode and to the customer's earth-continuity conductors. The resistance to earth of the electrode at the customers supply terminals is not specified but the overall resistance to earth of the LV system neutral must not exceed 20 ohms.

3.12.2 Pole- Mounted Substations (PNB)

If a supply is given to a single customer, PNB may be adopted.

The LV neutral conductor shall be earthed at the customer's supply terminals by connections to an earth electrode whose resistance to earth shall not exceed 20 ohms. The metallic sheaths of any LV cables shall also be connected to the neutral conductor at the customer's supply terminals. The transformer tank and associated HV metalwork will be connected to a HV earth electrode. All uninsulated parts of this electrode shall be at least 9 metres from any part of the LV earth electrode and any earthed metalwork connected to it.

Drawing C953072 shows typical arrangements.

The customer's installation shall comply with the conditions of Appendix A. If the LV network is subsequently extended, PME may be applied without affecting the existing customer, other than changing the PNB warning label.

Although it is preferable that the LV earth electrode is connected to the neutral conductor at the customer's supply terminals, the connection may be made at any other convenient position, for example on the route of under-eaves wiring.

3.13 Customer's Earthed-Sheath-Return Wiring Installation or Customer's Distribution Installation Using Combined Neutral and Earth Conductors

The ESQC Regulations (2002) permit the use of a PME terminal at the supply position provided that the network is suitable and the customers installation is bonded correctly. It does not provide a mechanism to enable the customer to use PME on an installation connected to the earth terminal.

3.14 Non-Standard Networks With inadequate neutrals

The construction of certain existing networks makes it impossible for Protective Multiple Earthing to be applied because in addition to the requirements for earthing at the substation

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	21 of 87

and on the network, there are conditions affecting the integrity and size of the neutral conductor stated in the ESQC regulations and in Engineering Recommendation G12 (Requirements for Application of PME to Low Voltage Networks) which CE have to meet.

The neutral conductor must not include a fusible cut-out, circuit breaker or switch between the supply transformer and any earth terminal connected to the neutral. If such devices exist on a network they shall be removed when multiple earthing is applied. (This does not preclude the provision of links in neutral conductors, but they shall be correctly installed and not removed while the phase conductors are live).

The copper equivalent cross-sectional area of the neutral conductor at any point should not be less than one half that of the phase conductor in a three phase, four wire or single-phase three wire main or service. (Two phase three wire systems include two phases of a three phase system). The copper equivalent cross-sectional area of the neutral conductor shall not be less than that of the phase conductor in a single-phase two wire main or service.

When using larger size three phase LV cables (typically 240mm² and 300mm²) for single phase applications consideration should be given to them having reduced size neutrals and not meeting the requirement for the neutral to be at least half of the size of the phase conductor. In cases where the larger size cable is installed to overcome volt drop limitations the smaller size neutral may be acceptable.

Where the neutral conductor does not comply with these requirements the network will have to be modified before PME can be applied. When such networks are identified they will be highlighted to Asset Management so that a work programme can be initiated for future replacement or upgrade of the network.

3.15. Customers Installations Requiring Special Attention

A PME earthing terminal provides a very satisfactory means of protection for the majority of customers installations but there are a number of special situations where the customer should provide an additional or alternative form of earth fault protection. The following situations fall into this special category:-

3.15.1 Construction Sites

PME earthing terminals shall not be provided on construction sites as it will not be possible to verify the requirements and conditions of Appendix A for bonding of the temporary water, gas and other metallic services including metallic structures which are in or might be expected to come into, electrical contact with earth and which are in easy reach of any exposed non-current-carrying metalwork of the Developer's or Contractor's electrical installation.

Construction site supplies shall be provided through a residual current device (RCD) which must be provided by the site Developer or Contractor. The construction site earth continuity conductor should be connected to an independent earth electrode which should be a minimum of 9 metres away from any HV earth electrode or cable sheath which may be on the site. The independent earth electrode should be of such a value that in the event of an earth fault, the maximum sustained voltage between the earth continuity conductor and the general mass of earth should not exceed the values prescribed in the current edition of BS7671.

Alternatively the Developer could install a 1:1 isolating transformer of the appropriate rating enabling a separately earthed neutral LV system to be created within the boundaries of the site.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	22 of 87

For large temporary supplies which require their own substation it will usually be possible to provide an earthing terminal connected directly to the transformer neutral in accordance with Section 3.12.1. For detailed requirements BS 7375 and BS 7671 should be consulted.

Where a CE PME earth terminal is provided for future use the earth terminal block shall be rendered inaccessible so as to prevent unauthorised connection. A notice with inscription "CE EARTH TERMINAL - NOT TO BE USED ON A TEMPORARY SUPPLY" shall be fixed adjacent to the earth terminal block.

3.15.2 Farms, Milking Parlours, Pig Stys etc

A PME earthing terminal shall be given to farm premises and similar establishments provided that the installation complies with the bonding requirements and conditions of Appendix A.

There are areas of special risk where animals could be subjected to a difference in potential between wet floors and bonded metalwork. In these situations the metalwork in the entire area of any farm buildings shall be bonded to the earthing terminal. If the customer wishes he may install a high sensitivity residual current device to provide additional protection in the event of a fault on the installation within the wet area. This is advisable where there is a bare concrete ground floor within the total area of the main buildings. The CE earthing terminal may be used as the circuit breaker earth electrode provided the whole of the metalwork within the main buildings is effectively bonded.

Where in remote buildings all extraneous conductive parts cannot be bonded to the earthing terminal, the pipes and metalwork of isolated buildings, whether or not they have an electricity supply, shall be segregated from metalwork connected to the PME earthing terminal. Any supplies to such buildings should be controlled by a RCD and the associated earth electrode and protective conductor shall be segregated from any metalwork connected to the PME earthing terminal.

Particular care must be taken in areas where livestock are housed as they are sensitive to very small voltages. A suitable metallic mesh shall be installed in the concrete bed of a dairy and bonded in accordance with the PME requirements.

If PME is to be applied to an existing dairy the steel reinforcement in the floor should be bonded. Alternatively if small voltage differences are unacceptable the area concerned should be protected by an RCD and the associated earthing system segregated electrically from the remainder of the installation.

(b) PNB Installations

Where a dedicated transformer is used to supply a single customer farm premises then PNB shall be adopted and shall comply with the conditions of Section 3.12.2 and Appendix A.

3.15.3 Swimming Pools

Where a PME earthing terminal is given with a supply to a swimming pool, including those at domestic premises, the customer shall be advised that the electrical supply to the pool should be separately protected with a high sensitive residual current device.

All metallic pipes and metalwork at the pool should be segregated from the metalwork connected to the PME earthing terminal and connected to a separate earth electrode. An

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	23 of 87

RCD shall be used to protect the supplies to the pool area. The length of insulated insert for metallic water pipes shall be as follows:-

Up to 25mm (1") diameter pipe 305mm (1'10")

Up to 50mm (2") diameter pipe 1.1 metres (3'6")

If segregation of the metallic pipes and metalwork is not practical then:-

All such metalwork including metal handrails and any other exposed metalwork which is in contact with earth shall be bonded together and an earth electrode installed.

An earth grid shall be installed around the pool and any associated shower areas and connected to the PME earth system.

A residual current device should be installed on the electrical supply to the pool.

It may be preferred that the lighting circuits are not protected by the RCD because of the possible danger presented by losing lighting when the pool is in use.

3.15.4 Sports Pavilions

Remote sports pavilions having PME supplies may present problems due to out of balance conditions, particularly if the service is of an appreciable length. Low values of voltage may appear on the neutral due to out of balance conditions. No problems will exist if concrete floors have an earth grid bonded to the earthing terminal. However, in the majority of cases the earth grid will not be installed and low levels of voltage could occur between the metalwork which is bonded to the neutral conductor and the general mass of earth (concrete floor) causing unpleasant sensations.

If an earth grid cannot be installed the customer should be advised to install a residual current device and the PME earthing facility should not be provided.

Where a sports pavilion is of timber construction, including the floor, then the difference in potential problem will not exist and a PME earthing terminal can be provided.

3.15.5 110V Centre-Tapped Supplies via an Isolating Transformer

A number of buildings with bar services have a 2kVA isolating transformer with a 110V centre-tapped secondary winding to give 55V to earth. A supply is given via a flexible armoured cable to delivery tankers which are fitted with 1hp pump motors.

The transformer does not however provide isolation for the centre-tap of the 110V winding if it is connected to the PME earthing terminal. The risk from providing supplies outside the equipotential zone is considered to be slight particularly as the time the tanker will be connected to the supply will be short.

3.15.6 Quarries

A PME earthing terminal may be provided for the part of the installation associated with permanent buildings such as offices provided that the building's electrical installation wiring conforms to BS 7671 and there is no interconnection with the earthing arrangements associated with the plant used in the production work of the quarry.

3.15.7 Petrol Filling Areas

The HSE Notes of Guidance for the provision of Petrol Filling Stations exclude the use of PME for the petrol filling area. PME can be provided for use in other buildings associated with the premises. The developer may request us to provide an SNE earth for the

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	24 of 87

development. If the supply is to be given from an SNE network then an SNE earth can be provided. The connection charge will include the additional costs of providing an SNE service.

If the service is to be supplied from a PME network then the developer will have to provide the necessary RCD protection for the petrol filling area.

3.15.8 Fairgrounds and Showgrounds

PME earth terminals will not be made available to such installations. A residual current device must be installed by the customer.

3.15.9 Railway Service Areas

It is recognised that under starting currents or traction fault conditions, a voltage gradient can appear on the traction return rail, which could be impressed on the supply cable. A PME earth terminal will only be provided where the railway operator can confirm that a voltage not in excess of 25V will be impressed on the distribution company earth terminal.

Note - Care is necessary with DC traction systems using earth return, as this can cause accelerated corrosion to exposed (i.e. Hessian served) cable sheaths and earth electrodes.

3.15.10 Caravans, Mobile Homes, Temporary Site Offices, Boat Installations and Outside Broadcast Vehicles

The Electricity Safety, Quality and Continuity (ESQC) Regulations 2002 do not permit the provision of PME earthing terminals for caravan and boat installations. This has the following implications:-

- (a) A supply may be given without an earth terminal to an individual caravan, caravan site, mobile home or temporary site office, from a PME distribution main.
- (b) A PME earthing terminal may be provided for use in permanent buildings on a caravan site such as the site owner's living premises and any bars, shops or wooden built toilet blocks provided that the electrical installation within the buildings meets the bonding requirements of the relevant British Standard.
- (c) A PME supply shall not be extended to shower blocks due to the higher probability of persons being barefooted at these locations unless a buried bonded metal grid has been installed. This requirement also applies to other locations where toilet or shower blocks have been provided for general public use where people are likely to be barefoot e.g. beachside locations, parks etc. Where outside showers have been provided then provision of PME earthing is not recommended as providing a reliable equipotential cage may prove impractical.
- (d) The site wiring shall be two-wire, phase and neutral, supplied through residual current devices, each one of which shall not supply more than 6 caravans. The residual current devices must be supplied by the customer or site owner.

The caravan installation should comprise separate phase, neutral and earth conductors. The earth conductor should be connected to an independent earth

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	25 of 87

electrode which may be common for up to 6 caravans. The earth electrode should be a minimum of 3 metres away from any PME earth electrode which may be on the site. The independent earth electrode should be of such a value that in the event of an earth fault, the maximum sustained voltage between the earth continuity conductor and the general mass of earth should not exceed 50 volts without the residual current device operating. For values of resistance of the independent earth electrode relating to various rated operating currents of residual current devices see Drawing C953027.

The main chassis of the caravan should be connected to the earth continuity conductor and hence to the independent earth electrode.

Drawing C953027 also shows the recommended method of supplying a caravan site and should also be used for yacht marinas and other similar installations such as sites with certain types of chalets.

- (e) A PME earthing terminal may be provided for use in mobile homes or temporary site offices which are constructed so that a person in contact with the general mass of earth cannot touch any metalwork of the home or site office which would be connected to a PME earthing terminal. Such mobile homes or site offices may be treated in the same manner as permanent premises and provided with a PME earthing terminal if the installation meets the bonding requirements of the relevant British Standard.

Where the mobile home or temporary site office has exterior metalwork and cannot comply with the above touch criterion, a PME earthing terminal shall not be provided. Supply should be given through a residual current device, and an independent earth electrode sited at least 3 metres away from any PME earth electrode should be provided. Both of these items are the responsibility of the customer.

- (f) A PME earthing terminal may no longer be provided for use in a vehicle used by the British Broadcasting Corporation, the Independent Broadcasting Authority or others for the purpose of outside broadcasts as the blanket approval is withdrawn. An RCD shall be provided by the customer.

3.15.11 High-Rise Buildings and Multi Occupancy Building Complexes

Engineering Recommendation G87 Guidelines for the Provision of Low Voltage Supplies to Multiple Occupancy Buildings should be read in conjunction with the following paragraphs which summaries the earthing requirements.

Networks within multiple occupancy buildings shall be designed on an SNE principle, although incoming service connections from local PME distributors are permissible, as is a PME earth terminal at the Intake Position. The appropriate main equipotential bonding connections to structural steelwork and to metallic services should be made at this point of connection, for details of the size of copper conductor required refer to appendix E. At individual Customer's Installations the main equipotential bonding between metallic services, extraneous metalwork and the earth terminal shall be carried out in accordance with BS 7671. This will ensure that no harmful potentials appear between earthed and extraneous metalwork within the Customer's premises under fault conditions. A single Intake Position is the preferred design option for, amongst other reasons, avoiding

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	26 of 87

problems caused by the flow of neutral current through the building steelwork and equipotential bonding between the Intake Positions. Note that a single Intake Position may have more than one incoming service cable and cutout. If it is not possible to design on the basis of a single Intake Position, then a non-PME connection option shall be considered (ie shall be of the SNE type or no earth shall be provided from the network the landlord making their own earth provision). All Intake Positions within a single building shall employ the same earthing method. PME connections are not recommended in these situations due to the risks associated with significant neutral current flow through structural steelwork and fixings under fault or other conditions and the difficulty associated with providing an alternative DNO-owned bond between Intake Positions.

In some steel framed multiple occupancy buildings such as a horizontal run of industrial or commercial units where individual services are to be provided from a DNO's PME distribution main it will be more difficult to avoid the problems identified earlier. There are several ways of addressing this which will vary to the extent that they are able to mitigate the problem.

- Provide a single Intake Position to the building with a group metering position at a communal point. A Customer's separate neutral and earth cable could then be supplied to each unit separately.
- Operate all Customer's Installations on the same site as TT systems where practicable (ie no DNO earth terminal provided). This solution has the advantage that it eliminates the problem but may not be suitable for all load types.
- Provide a PME connection to a freestanding distribution pillar and then distribute with separate neutral and earth cables in a star formation from this point to each unit. Note that the neutral and earth conductors of the cables **MUST NOT** be bonded together at any point other than the pillar as this would result in a path for neutral current to be diverted into the building structure.

Individual PME connections are not recommended in these situations due to the risks associated with significant neutral current flow through structural steelwork and fixings under fault or other conditions and the difficulty associated with providing an alternative DNO owned bond between Intake Positions. In the case of multiple PME cutouts at a single Intake Position, all earth terminals shall be bonded together. The size of the bonding conductor shall be not less than that of the smallest associated service cable neutral conductor. Such bonding conductors shall be coloured green-yellow with the addition of blue markers to indicate that they will be carrying current.

Drawing C953070 shows the bonding requirements for an SNE rising main.

Where minor changes or extensions are carried out to existing complexes which have been supplied with CNE mains safeguards have to be taken with regard to the connection of the supply neutral conductor to earth at the extremities of the LV distribution systems in the building complexes. It is essential that the PME earth shall provide a continuous metallic path from the remote end of the distributor to the neutral at the substation.

The connections at the various ends of the neutral conductors shall be made in one or more of the following ways:-

- (a) On a ring system the neutrals are made continuous at normally open points.
- (b) A bonding lead connecting the neutrals between the remote ends of two

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	27 of 87

distributors.

- (c) A separate earth conductor run from the remote ends of the distributors to an earth electrode or the neutral at the substation.

(Note - Potential EMC problems may occur particularly if there is more than one PME supply cable, where neutral return currents could flow through the metal frame of the building appendix C provides guidance on minimising EMC problems)

Incoming gas and water services and structural steel work in contact with the general mass of the earth will be bonded to the earth bar of the incoming distribution panel in accordance with Appendix A.

3.15.12 Supply Terminating in a Separate Building

Occasionally a service will terminate in a position remote from the building it supplies. In this case the size of the PME bonding in the building supplied must be related to the size of the incoming supply cable in accordance with appendix E. If the size of the circuit protective conductor of the cable between the supply intake position and the building is less than that of the PME bonding conductor, a suitable additional conductor will be installed.

3.15.13 PME and Outside Water Taps

Under an open-circuit supply neutral condition the potential of an outside water tap will rise above earth potential. A person coming into contact with the tap could receive an electric shock and the shock could be severe if that person were barefooted. The probability of these two conditions occurring together is considered to be so small that the use of PME where a metal outside tap exists is not precluded.

An insulating insert may be incorporated in the pipe to the outside water tap. However care should be taken to ensure that simultaneous contact with metal pipework on each side of the insert is not possible if there is likely to be a potential across the insert under this condition

3.15.14 Lightning Protection Systems

For the majority of situations it is acceptable for lightning protection to be connected to the customer's earthing arrangements providing it has been designed and installed in accordance with the latest version of BS 62035 "Protection against Lightning".

When periodic testing of the lightning protection system is undertaken, precautions are necessary when breaking the link between the lightning protection electrode and the customer's earth since the customer's earth may be at a potential above true earth.

3.15.15 Roadside and Other Housings Accessible to the Public

Examples are Public Telephones, Pedestrian Crossing Bollards, Ticket Machines etc. Equipment in this type of structure should be Class II or equivalent construction. No mains-derived earthing terminal is required, neither is a residual current device needed for earth fault protection.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	28 of 87

The service arrangement to BT public telephone housings is covered in Engineering Recommendation PO4/1. It is suggested that telephone housings owned by other telephone operators should have similar service arrangements.

Other freestanding structures not of Class II construction such as small brick or metal housings, or metal cabinets containing pumps, motors, controls etc, may be offered the use of a PME terminal provided the supply is single phase and the maximum load does not exceed 2kW. An earth electrode with a value not exceeding 20ohms should be supplied and installed by the customer and connected to the earth bar.

If these conditions are not met, a PME terminal should not be offered. No earth terminal shall be made available from the network, the customer will install their own earth and protect the circuit by installing an RCD.

For other wall or pedestal mounted structures not of Class II construction which require a single phase supply, the use of a PME terminal should not be offered. No earth terminal shall be made available from the network, the customer will install their own earth and protect the circuit by installing an RCD

For installations requiring three phase supplies where the load is balanced, PME can normally be offered.

Crash barriers adjacent to street furniture should not be connected to a PME earth terminal

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	29 of 87

3.16 Under-Eaves Wiring Installations

3.16.1 New Under-Eaves Wiring Installations

PME shall be adopted on all new under-eaves wiring installations which will be installed in accordance with NSP/004/003 Specification for Overhead Services, Surface Wiring and Eaves Wall Mains. Obligatory connections of the neutral conductor to earth electrodes or neutral conductors of other distributors shall be made at the final insulated end on underground networks and at the final support on overhead networks of every branch of the distributor.

Under-eaves wiring installations supplying 5 or more adjacent customers will be deemed to be a branch of a distributor and the neutral conductor shall therefore be connected to an earth electrode at or near the last service termination on the branch.

Under-eaves wiring installations supplying up to 4 adjacent customers need not be considered as a branch of the distributor for this purpose. However, the following exceptions will apply:-

1. Where an overhead service exceeds 3 spans in length an additional earth electrode shall be installed either on the overhead-service termination where the connection is made to the under-eaves wiring installation or at the final support of the overhead line.
2. Where an under-eaves wiring installation includes an overhead interhouse span between customers then an additional earth electrode shall be installed either at the most remote service of the under-eaves wiring or at the termination of that overhead span which is most remote from the substation. If the overhead span consists of a CNE concentric cable or ABC then an additional earth electrode need not be installed.

3.16.2 Existing Under-Eaves Wiring Installations

Earthing arrangements of underground networks with electrically continuous sheaths are satisfactory and those with discontinuous sheaths are unsatisfactory; the latter shall be brought up to the required standard as described in Section 3.8.2.

Earthing arrangements of non PME overhead CE networks without aerial earth-wires are unsatisfactory and must be converted to PME. Overhead networks with aerial earth-wires are acceptable in the short term and can be converted to PME.

Under-eaves wiring installations without an earth conductor shall be either converted to PME or a separate earth conductor will be installed whichever is the more economical. The earth conductor shall be bonded to the sheath of the supply cable of the underground cable network or to the aerial earth wire of the overhead network.

Where the under-eaves wiring installation is connected to an overhead network without an aerial earth wire the customers on that installation shall be converted to PME.

Where the under-eaves wiring installation is converted to PME, 5 or more adjacent customers will be deemed to be a branch of a distributor and the neutral conductor shall therefore be connected to an earth electrode at or near the last service termination on the branch.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	30 of 87

Under-eaves wiring installations supplying up to 4 adjacent customers need not be considered as a branch of the distributor for this purpose. However the following exceptions will apply in both the 4 customer case and the case of 5 or more customers.

Where an under-eaves wiring installation is converted to PME or where a separate earth conductor is installed and the under-eaves wiring installation includes an overhead interhouse span between customers then an additional earth electrode shall be installed either at the most remote service of the under-eaves wiring or at the termination of that overhead span which is the most remote from the substation. This additional earth electrode shall be connected to the neutral conductor in the case of PME or the earth conductor where one is installed. If the overhead span consists of a CNE concentric cable then an additional earth electrode need not be installed.

Where an overhead service exceeds 3 spans in length an additional earth electrode shall be installed on the overhead service termination where the connection is made to the under-eaves wiring installation or at the final support of the overhead line.

Drawing C953074 shows typical arrangements of PME earths on under-eaves wiring installations.

3.17 Customers' Private Generator Installations

Recommendations for the connection of small scale embedded generators can be found in Engineering Recommendation G83. For larger embedded generators with a capacity greater than 16A per phase the requirements are covered by Engineering Recommendation G59

3.17.1 PME/PNB Installations

Where private generators are used to provide standby supplies on PME or PNB installations a three-pole (as opposed to a four pole) changeover switch must be used. This is because on a PME or PNB installations the neutral conductor cannot be disconnected simply by the breaking of the neutral conductor in the changeover switch since the earth connection to the generator neutral conductor and the bonding of this same earth to the earth terminal block and neutral at the service position prevent the isolation of the generator neutral from the supply cable.

3.17.2 Non-PME Installations

The connection of a generator to the permanent wiring installation in a customer's premises must include a changeover switch which will completely isolate all phases and neutral of either the permanent or the standby generator supplies. The purpose of the switch is to prevent interconnection of the supplies and the possibility of "feedback" from the generator into the CE supply mains.

The generator neutral terminal shall be bonded to the frame earth terminal and the neutral conductor of the installation.

The earth terminal of the generator shall be connected to an earth electrode which must be provided by the customer as close as practicable to the generator. The resistance to earth of this earth electrode shall not exceed 20 ohms. In addition, the earthing terminal of the generator should be connected to the earthing terminal block at the CE intake unit. In the case of large factory installations, the earth terminal of the generator may be connected to the largest cross section earth conductor at the distribution point, instead of earth terminal at the CE intake unit.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	31 of 87

3.18 Connections to Mobile Phone Base Stations associated with Transmission Towers or Primary Substations

Under high voltage fault conditions exceptionally high earth potential rise (EPR) conditions can occur on transmission towers and at some primary substations. Steps must be taken to avoid applying these voltages to the local LV and 11kV systems. Consequently site specific studies and designs are required to ensure the safety of connected customers and network operatives when providing an LV connection. The current edition of Engineering Recommendation G78 gives recommendations for the provision of LV connections to mobile phone base stations with antennae on high voltage structures DSS/022/001 Code of Practice for Low Voltage Connections to Mobile Telephone Base Stations with Antennae on High Voltage Structures is based on G78 and should be referred to.

3.19 Cathodic Protection Installations.

The usual source of power for cathodic protection installations is mains supplied transformer rectifier unit.

The preferred arrangement is for the customer to provide his own earthing together with an appropriate Residual Current Device. An SNE or PME earth can be given provided there is no electrical connection between the primary and secondary of the transformer rectifier unit.

Where supply is given via a pole-mounted transformer the substation earth electrodes and stay wires should be situated as far as possible from the ground bed.

BS 7361 gives more information on cathodic protection.

3.20 Earthing at Distribution Substations

Earthing at all CE 6kV, 11kV and 20kV distribution substations shall satisfy the following requirements:

3.20.1 Earthing requirements at new distribution ground mounted substations

3.20.1.1 Introduction

This section summarises the standard arrangements for the earthing and bonding of the most common standard types of 20kV, 11kV and 6kV Ground-Mounted Distribution substations owned and installed by CE.

Three common standard arrangements are provided in section 3.20. Where an HV substation is to be sited within or adjacent to, a higher voltage substation (e.g. 132/33kV, 33/11kV, etc), the guidance of the Design Engineer must ALWAYS be sought before proceeding with the construction process.

3.20.1.2 General Earthing Requirements

For a fault directly to earth or between HV and LV transformer windings, a path is required to allow sufficient earth fault current to flow back to its source, operate protection and disconnect the supply. An important part of the path is the earth electrode at the substation. For the majority of ground mounted substations, a combined HV/LV earth with will be installed. (See provisos below)

The electrode will consist of bare conductor and earth rods installed around the HV equipment and bonded to the substation equipment.

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 32 of 87

Exposed, earthed metalwork associated with HV plant (transformer, switchgear etc) is to be connected to the bare electrode in and around the substation. The LV transformer neutral is to be connected to the same electrode, or if separately earthed via insulated conductor to the LV electrode system situated at least 9m away from the substation.

Where the HV and LV earths are combined an electrode will be laid for a minimum of 80m out with incoming and outgoing HV cable routes. Where a new substation is teed into an existing network the additional electrode will be laid the length of the HV trench where this is less than 80m.

Bonding of the HV and LV earths can be achieved using a combined electrode or the fully rated link within the LV distribution cabinet, which connects the LV neutral bar to the HV switchgear earth bar.

Where the systems are combined, the neutral-earth connection and LV earth electrode will form part of the return path for HV fault current.

The HV and LV electrode systems can be combined only where the following two conditions apply:-

- (i) The conditions in paragraph 3.20.5 are met (the substation is connected to a global earthing system).
- (ii) The EPR at the substation is less than 430V during fault conditions.

The conditions can normally be satisfied if the substation is in an urban location, within an all underground cable network and not the first substation on a circuit out from a primary substation.

Where the HV and LV earths are not combined because the conditions (i) and (ii) above can not be met (normally substations teed off an overhead line) earth resistance values not exceeding those given in table 3.20.1.2 will be achieved. Separate HV and LV electrode systems are required which are electrically segregated via a 9m separation in the soil, in these situations no additional electrode will be run with the outgoing HV cable routes.

It is the company's policy to identify substations where HV and LV electrodes are not combined, by fixing a label to the outside of the LV cabinet.

Where HV and LV earth electrodes are segregated the following maximum values shall apply:-

Table 3.20.1.2

System Voltage kV	Maximum Value	
	HV Earth	LV Earth
11	20 Ω	20 Ω
20	20 Ω	20 Ω

3.20.2 Bonding and Control of Potential

3.20.2.1. General.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	33 of 87

(i) Bonding of HV and LV Electrodes.

At new ground mounted substations in urban areas supplied from global underground networks, the HV and LV electrode systems will normally be combined, i.e. a separate LV electrode is not necessary. In all other areas, separate HV and LV electrode systems shall be installed. Where these are kept separate, a warning notice is to be fitted and shall have the following wording:-

CAUTION

Avoid making simultaneous contact between the high voltage metalwork earthing and low voltage earthing systems.

3.20.2.2 Bonding.

All normally accessible metalwork will be connected together and to the HV earth, via bonding conductors. This is to ensure that all adjacent, exposed metalwork remains at a similar potential during fault conditions. The bonding conductor between main items of plant and the electrode system could carry significant earth fault current and must be fully suitable for this purpose see appendix E.

Exposed metalwork (transformer and switchgear enclosures, metal fencing, metal doors, metal enclosures, LV enclosures) will be bonded together and to the main HV earth electrode (bare electrode in and around the substation).

Doors, metal roofs etc., will not carry appreciable fault current, other than via a secondary wiring fault, so smaller bonding conductors are permitted. In all cases the minimum size is 16mm² copper or 35mm² aluminium.

3.20.2.3 Control of Potential.

During an earth fault, protection must be provided for anyone who could be in contact with exposed earthed metalwork. This applies particularly to those required to carry out fault switching at the substation. This protection can be provided by keeping the operators hands and feet at a similar potential.

This is achieved by laying bare HV electrode, which is connected to exposed metalwork and switchgear, in the ground such that it is immediately beneath and just beyond the position where the operator stands

The following general conditions apply:

For ground mounted substations supplied from an overhead network or substantive underground network, a standard arrangement which provides potential grading and a reasonably low resistance, is always required. The standard earthing arrangements in section 3.20.7 should suffice without any further additional electrode.

3.20.3 Distribution Substations near or within a Higher Voltage Substation.

Such cases are to be referred to the Design Engineer.

3.20.4 Customer Substations Supplied at HV.

Wherever possible, one of the standard arrangements having a combined electrode will be used. The HV earth will be installed by CE, whilst the LV electrode will be the responsibility

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	34 of 87

of the customer to design, install and maintain. Where the systems have to be segregated, a nine metre separation is required between the LV electrode and any HV electrode or exposed HV metalwork.

Where the substation forms part of a larger building it may not be possible to install an earth electrode around the full perimeter of the substation. In these cases it is permissible to install the electrode 1 metre out from the external walls. Other than the use of a partial perimeter electrode all other bonding and installation details contained in the standard design shall be followed.

Should the customer wish to connect the LV electrode to the HV one, the matter is to be referred to the Design Engineer.

The Engineer responsible for site works during construction is required to confirm that the HV and LV electrode (if to be combined) is installed to the company's specifications. Testing of the earth value and, if appropriate, for segregation, should be carried out prior to commissioning, following the approved CE procedure.

Where a standard arrangement is not possible and the customer has HV equipment/plant, then two HV electrode systems will be required. CE will install its own, generally consisting of bare electrode within the incoming cable trenches. The customer will install a system which, as a minimum, encircles their equipment/plant on all sides, or those sides not covered by the company's electrode. Ideally, the two HV earthing systems will be designed as one entity and be combined to provide a low overall earth impedance. If not designed together, then before a connection between them is permitted, the Design Engineer must ensure that the combined earthing system will be satisfactory and agree the method of connection.

Where the substation forms part of a metal framed building which requires bonding between the LV earth and the building structure, to satisfy the IEE wiring regulations BS7671 and there is a need to segregate the HV and LV earth to satisfy the ESQC Regulations the HV earth connection will be insulated. The HV earth electrode will be 20Ω or less and installed insulated from the LV earth electrode and metal structure of the building by 9 metres. Simultaneous contact must be avoided between the HV and LV earth systems in these cases the most practical solution may be to install the HV switchgear and transformer in a separate room or screened off area to prevent contact between the earthed HV metalwork and the LV distribution equipment.

3.20.5 Combined HV/LV Earths and the Global Earth Concept

A global earth system is an equivalent earthing system created by the interconnection of local earthing systems made up of earth electrodes and cables with metallic sheaths in direct contact with earth that ensures, by the proximity of the earthing systems, that there are no dangerous

touch voltages. Such systems permit the division of the earth fault current in a way that results in a reduction of the earth potential rise at the local earthing system. An example of a global earth system is one associated with a city centres where the majority of cables have metallic sheaths in direct contact with earth.

In urban situations it is desirable that ground mounted distribution substation HV and LV earths are bonded together to achieve the lowest practical earth value in an environment where installation and effective segregation of earths is often difficult or impossible. Previously the only criterion that had to be met was the combined values of the high and low voltage earths should be 1ohm or less this is no longer acceptable.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	35 of 87

If HV and LV earths are to be combined there is an obligation for CE to ensure the potential on the combined HV/LV earth electrode and in turn customers exposed conductive metalwork does not rise to unacceptable levels (above 430V) under fault conditions. The two main fault scenarios are:

- HV faults at or adjacent to the distribution substation which may or may not be connected to it associated primary substation by a continuous metallic cable sheath.
- EHV faults at the source primary substation causing the primary substation and its HV cable system to become 'Hot'.

The possibility of the earth potential rise (EPR) on the combined HV/LV earth electrode rising above 430V can generally be ignored when:-

- 1) For distribution substation locations **with NO** continuous metallic sheath connection back to the source primary substation (i.e. there is a section of overhead line in the circuit between the source substation and the distribution substation) where all of the following criteria listed below can be met :
 - i. The distribution substation concerned is located in an area with a uniform soil resistivity of less than 200Ω m.
 - ii. A gross source 11kV earth fault current of less than 2500A.
 - iii. At least 8kM of metallic sheathed cable in direct contact with the soil.
 - iv. The distribution substation concerned will also have metallic interconnection via HV or LV cable sheaths to at least one other distribution substation.

OR

- 2) For distribution substation locations **with a** continuous metallic sheath connection back to the source primary substation where all of the following criteria listed below can be met :
 - i. A continuous metallic sheath connection back to the source primary substation bonded effectively at each substation.
 - ii. The distribution substation concerned is located in an area with a uniform soil resistivity of less than 200Ω m.
 - iii. A gross source 11kV earth fault current of less than 2500A.
 - iv. At least 2kM of metallic sheathed cable in direct contact with the soil with a continuous cable sheath connection back to the source primary substation.
 - v. The distribution substation concerned will also have metallic interconnection via HV or LV cable sheaths to at least one other distribution substation.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	36 of 87

- 3) In addition to the requirements of 2) above the primary substation supplying the 20kV or 11kV underground network is connected back to its 132kV or NGT source via underground cables with continuous metallic sheaths.

If the above conditions are met the HV and LV earths of urban ground mounted distribution substation may be bonded together.

If any of these criteria cannot be met further investigation will be required.

- If condition 1 or 2 cannot be met the actual earth return current at the proposed distribution substation can be calculated and Tables 1 and 2 consulted using this current in place of the source fault current.
- If condition 3 cannot be met refer to the NEDL/YEDL Earthing Databases (It may show that a primary substation is cold).

If after this further investigation the requirements for combining the HV and LV earth can not be met the earths shall be separated with reference to the guidance in section 3.20.7.3

Table 1

Calculated Impedance of PILCSWA Cable Networks with metallic sheaths in contact with the soil - Assumes No Continuous Metallic Sheath Path Back To Source

Resultant EPR's > 430V are shown highlighted in grey

Total* Length of PILCSWA cable - kM	Impedance ohms	EPR for Gross Source Earth Fault Currents of:					
		3000A	2500A	2000A	1500A	1000A	500A
2	0.431	1293	1077.5	862	646.5	431	215.5
4	0.247	741	617.5	494	370.5	247	123.5
6	0.188	564	470	376	282	188	94
8	0.155	465	387.5	310	232.5	155	77.5
10	0.136	408	340	272	204	136	68

*Metallic cable sheaths must be in direct contact with the soil.(hessian serving allowed)

Cable in ducts or with plastic sheath do NOT count.

Cables following parallel routes do NOT count.

Metallic sheathed LV cables may only be counted if they are connected to a common HV/LV earthing system

Uniform Soil Structure 200Ωm

Cable size used for modeling 0.1 sq in 11kV

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	37 of 87

Table 2

Calculated Impedance of PILCSWA Cable Networks with metallic sheaths in contact with the soil. Assumes Continuous Metallic Sheath back to source substation (maximum 65% reduction - minimum 35% ground return)

Resultant EPR's > 430V are shown highlighted in grey

Total* Length of PILCSWA cable - KM	Impedance ohms	EPR for Gross Source Earth Fault Currents of:					
		3000A	2500A	2000A	1500A	1000A	500A
2	0.431	453	377	302	226	151	75
4	0.247	259	216	173	130	86	43
6	0.188	197	165	132	99	66	33
8	0.155	163	136	109	81	54	27
10	0.136	143	119	95	71	48	24

*Metallic cable sheaths must be in direct contact with the soil.(hessian serving allowed)

Assumes continuous Sheath back to source primary substation.

Cable in ducts or with plastic sheaths do NOT count.

Cables following parallel routes do NOT count.

Metallic sheathed LV cables may only be counted if they are connected to a common HV/LV earthing system

Uniform Soil Structure 200Ωm

Cable size used for modeling 0.1 sq in 11kV

The following issues/cases should also be considered: -

- 1) If the substation is connected to a modern HV cable network with all insulated cable sheaths, e.g. PICAS, XLPE etc. In this case the HV and LV cable sheaths will make no significant contribution to the HV earthing system. (They are generally the most important factor in lowering earthing resistance in older systems with hessian served metallic sheathed cables) Unless carefully managed there is a natural tendency for such systems to have high resistances to earth and potential rises under fault conditions. The benefits of Global earthing tend to be lost overtime as more insulated sheath HV and LV cable is installed and cables with metallic sheaths in contact with the soil are removed.
- 2) In order to maintain a global earthing system the following action will be taken: -
 - o Often reinforcement schemes involve replacement of lead sheathed, steel armoured 6.6, 11 or 33kV cables. The phase conductors and sheaths of these old cables, once disconnected may be joined together and they shall be connected to the electrode system. Due to the high risk of damage, it is

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	38 of 87

essential that multiple connections are provided to such cables. They should also be marked up on the cable records as supplementary earth electrodes. The remote ends (or intermediate points) must also be connected to the electrode system by connections to the cable sheath at suitable points, such as at joints or distribution substations.

- o A bare copper stranded earth electrode of 70mm² cross-sectional area shall be laid with each out-going group of insulated sheath cables for a minimum distance of 80 metres from the substation. Care should be taken to avoid placing this electrode in areas subject to a rise of earth potential from adjacent towers or EHV substations.
 - o In newly developing urban areas where a high proportion or possibly all the network has or will be plastic sheathed HV and LV cables and where there is higher than average soil resistivity (>200Ωm) consideration will be given at the network design stage to installing addition earth electrode in excess of the 80m mentioned above at each distribution substation in such cases guidance of the policy manager for earthing will be sort.
- 3) The use of land in which bare earth electrodes are placed should be considered. Locations to avoid include sites where people may be expected to walk barefoot such as caravan sites and pathways to beaches etc. Animals have a relatively low step voltage (25V), hence routes where farm animals may be expected to pass or congregate frequently such as gateways and drinking/feeding troughs etc should be avoided.
- 4) Bare copper conductor can corrode in hostile soil environments causing earthing systems to become ineffective. Soils containing sulphates, sulphides and acids should be avoided.
- 5) HV cables with insulated sheaths or where the sheath is not in close contact with earth will not contribute to the global earth. Examples are XLPE, aluminium - or lead-sheathed cables with PVC oversheath, and cables laid in bitumen or compound-filled troughing or pipes, or laid on cable trays. Hessian or jute serving is not considered to form an insulated sheath.
- 6) It will generally be preferable to adopt a combined HV/LV earthing arrangement at existing substations where lead-sheathed HV and LV cables have been installed. The alternative, i.e. electrical separation of the two electrode systems, would involve the excavation and insulation of the LV cable sheaths, either by wrapping with a non-hygroscopic insulating tape, enclosing in an insulating duct or replacing by CNE cables. At new substations where CNE cables are used the LV neutral may be readily isolated from the HV earth electrode at the substation and earthed via a separate LV earth electrode.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	39 of 87

3.20.6 Construction Details

3.20.6.1 Materials.

The standard materials are:

- (i) Electrode conductor - hard-drawn copper conductor of 70 mm² or 0.1 inches². (This can cater for up to 8kA fault current for three seconds, as required by ENA TS. 41-24).
- (ii) Extensible earth rods - copper clad to ENA TS 43-94. 1.2m long 16mm diameter and / or deep driven 70mm² stranded copper.
- (iii) Bonding conductor for connecting roof and metal walls of external enclosure - stranded copper, PVC covered, flexible 16mm² cross sectional area.
- (iv) Inter-plant bonding conductor - stranded, copper conductor of 70mm² cross section, PVC covered, to BS6004.
- (v) Flexible copper connecting braid of 16mm² minimum.
- (vi) Flexible copper power cable gland connections of minimum 50mm².

3.20.6.2 Installation of Electrode.

The standard electrode arrangement will be installed during the civil work associated with construction of the substation. Where it can be reasonably assumed that the soil resistivity will be greater than 200Ωm (i.e. in arrears where the soil is made up primarily of slate, shale or rock), prior to, or during the civil works, soil resistivity readings should be taken. Once interpreted, these can be used with tables 3.20.7.2 and 3.20.7.3 to provide an estimate of the resistance value. If the calculated resistance is too high, the guidance in the tables should be followed to reduce it. Normally this is by installing longer vertical rods. Alternatively, or in addition, earth electrode can be installed in one of the HV cable trenches, provided this is 9m from any planned LV electrode.

The LV electrode can be laid in an LV cable trench, providing that 9m separation is maintained between this electrode and any HV metalwork, HV hessian served cables and the HV electrode. Guidance is provided in Table 3.20.7.3 concerning the amount of electrode required to achieve the necessary LV resistance value.

Care must be exercised when driving earth rods because of the possibility of puncturing / damaging cables or other services buried below the cable trench level. This is especially important when rods are installed after the completion of cable laying. Cable locating tools and techniques must be used in addition to consulting accurate cable records, to avoid damaging such cables/services.

Where earth conductors are laid adjacent to HV cables, the backfilling medium will be dictated by the cable thermal requirements. It is standard practice to lay the earth conductor on the bottom of the trench, below the cables, so it is in direct contact with the surrounding soil. Ideally there should be 0.5m clearance between hessian-covered or plain tape-armoured conductors and the earthwire, but this can be reduced to 0.2m when the cable or earthwire has an insulated outer sheath.

Horizontal electrode adjacent to the substation door (and any other earth electrodes forming part of measures to reduce the effect of touch voltage) are to be bedded in suitable backfill, then covered with a layer of crushed rock or chippings or concrete slabs.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	40 of 87

Under no circumstances are earth electrodes to be installed in coal ash (or similar material), which is normally very corrosive to copper.

3.20.6.3 20/11/6.6kV Cables

The earth sheaths (and armours, if present) are to be terminated via a flexible connection onto the switchgear earth bar that interconnects the transformer and HV switchgear. The flexible connection shall be sufficiently long to accommodate earth passage fault indicators or automation transmitters and shall be of 50mm² (minimum) copper. It is important that these connections are of low resistance and secure against loosening by vibration etc. The resistance of the connection and joint is to be measured using a micro-ohmmeter and will be below 100μΩ.

3.20.7 Standard Arrangements

Construction layouts have been produced to cover the main types of new HV substation, including kiosk type,. The layouts have been designed to meet the following requirements:

- (i) HV and LV earthing systems are installed separate or bonded.
- (ii) HV and LV electrodes if separated will each have a maximum value of 20Ω.
- (iii) Full bonding within the substation is provided.
- (iv) Additional protection is provided to protect operators, staff and the public against possible excessive voltage rise, where applicable.

Standard construction drawings, tables which give the S/S electrode resistance and where necessary a separate LV electrode resistance for a range of uniform soil conditions and notes of guidance are given for three typical ground mounted substation designs.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	41 of 87

3.20.7.1 Standard Arrangement Drawings.

The following arrangement drawings provide details of the construction practice necessary. Reduced size drawings are included at the end of this section, but for construction purposes, the current up-to-date versions of the drawings should be consulted.

Substation Arrangement	Earthing Arrangement	Drawing Number	Arrangement No.
Close Coupled Ground Mounted Substation.	Combined HV & LV Earth	C944571	1
Close Coupled Ground Mounted Substation.	Separate HV & LV Earth	C951997	2
Steel Kiosk Type Ground Mounted Substation.	Combined HV & LV Earth	C944570	3
Separate HV Switchgear Ground Mounted 20kV Substation.	Combined HV & LV Earth	C944575	4

3.20.7.2 Standard Arrangement Resistance Values

The above standard arrangements produce the following resistance values in different soil types:-

Soil Type	Typical Soil Resistivity (Ωm)	Resistance of Standard Earthing Designs with 1.2m Earth Rods at S/S*			
		Arrangement 1	Arrangement 2	Arrangement 3	Arrangement 4
Loam	25 or less	2.2 Ω	2.1 Ω	1.7 Ω	2.2 Ω
Chalk	50 or less	4.4 Ω	4.2 Ω	3.3 Ω	4.3 Ω
Clay	100 or less	8.7 Ω	8.3 Ω	6.5 Ω	8.5 Ω
Sand/Gravel/Clay mix	150 or less	13.0 Ω	12.4 Ω	9.7 Ω	12.7 Ω
Sand/Gravel/Clay mix	200 or less	17.3 Ω	16.6 Ω	12.9 Ω	16.9 Ω
Slate/Shale/Rock mix	500	43.2 Ω	41.4 Ω	32.2 Ω	42.2 Ω

*2.4m rods required by CE at rural locations.

Where the HV and LV earth are separated additional HV earth electrode to the standard design will be required for soil resistivities in excess of around 200 Ωm to maintain a 20 Ω maximum HV earth resistance value.

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 42 of 87

3.20.7.3 Separate LV Earth

Where it is necessary to install a separate LV earth with a maximum 20Ω value then the following alternative arrangements shown in the table below may be installed:-

Arrangement 1 gives the required length of horizontally laid bare conductor (A minimum length of 5m is suggested) required to achieve an LV electrode value of less than 20Ω for various soil resistivities

Arrangement 2 is a simple horizontally laid conductor installed in incremental lengths of 5m plus a 3m vertical rod at 5m intervals.

Bare LV earth conductor shall be separated by at least 9m of insulated conductor from the nearest part of the HV electrode system. (This includes non plastic sheathed HV cable such as PILCSWA cables with hessian serving).

Soil Resistivity (Ωm)	LV Electrode Arrangements to give 20Ω	
	Arrangement 1 Horizontal Electrode Only (Length Required)	Arrangement 2 3m Vertical Rods + 5m Horizontal Electrode Segments
25	5m	
50	6m	
100	12m	
150	20m	
200	28m	
500	75m	3m Rod x 7 plus 7 x 5m = 35m Horizontal
		Key Key 3m Vertical Rod   5m Horizontal Electrode

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	43 of 87

3.20.7.4 Standard Earthing Arrangements for non metallic substations

HV Electrode

Perimeter electrode installed on the outer edges of the foundation, in direct contact with the soil at a depth of 1.2m.

Internal cross conductor installed along the outer edges of the cable trench at 0.9m depth and connected at each end to the perimeter electrode.

Two connections from the internal cross conductor brought up to the switchgear earth bar and fitted adjacent to one another.

Four off 1.2m earth rods installed on the corners (2.4m rods in rural areas).

If metal doors are used, then additional grading electrode will be required in front of the doors.

Copper earth tape 25 x 4mm.

Bonding

All main items of plant must be bonded to the main switchgear bonding bar with either 25 x 4 mm copper tape, 25 x 6mm aluminium tape or 70mm² PVC covered copper. The bonding bar and bonds to each main item of plant are normally supplied installed on new plant by the manufacturer. Connections to metalwork must be free from paint, cleaned and a protective coating of grease applied.

LV Electrode

Not required where the HV and LV earths are to be bonded, as the HV earth will fulfil the necessary requirements.

If separate earths are required (see para. 3.20.7.3 and Table) a minimum distance of 9m must be maintained between this electrode and any 11kV electrode, the substation compound and any other metalwork connected to the HV earth. The LV electrode will be connected to the LV neutral busbar in the LV cabinet via at least 9m of 70mm² PVC covered copper conductor. The PVC conductor must be compression bonded to the LV earth electrode at one end, and to the neutral bar of the distribution cabinet at the other, via CE approved connectors.

High Voltage Cables

The outer metal sheath of 11kV cables will be bonded to the switchgear bonding bar via a flexible strip or braid having a current carrying capacity equivalent to at least 50mm² copper. This is to be of sufficient length to accommodate a fault current indicator.

Low Voltage Cables

For modern CNE cables, the outer sheath of the cable will be connected to the neutral bar in the distribution pillar using a CE approved compression or mechanical connector.

For SNE cables the outer sheath and armouring should be bonded together and connected to the earth bar in the distribution pillar. The neutral conductor is then connected to the neutral bar.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	44 of 87

3.20.7.5 Standard Earthing Arrangement for Steel Kiosk Substations.

HV Electrode

Perimeter electrode installed on the outer edges of the foundation, in direct contact with the soil at a depth of 1.2m.

A second perimeter electrode laid around the substation, 0.6m deep and up to 1m away from the foundation.

Internal cross conductor installed along the outer edges of the cable trench at 0.9m depth and connected at each end to the perimeter electrode.

Two connections from the internal cross conductor brought up to the switchgear earth bar and fitted adjacent to one another.

Two off 1.2m earth rods installed on the internal perimeter electrode (2.4m rods in rural areas).

Copper earth tape 25 x 4mm.

Bonding

All main items of plant must be bonded to the main switchgear bonding bar with either 25 x 4 mm copper tape, 25 x 6mm aluminium tape or 70mm² PVC covered copper. The bonding bar and bonds to each main item of plant are normally supplied installed on new plant by the manufacturer. Connections to metalwork must be free from paint, cleaned and a protective coating of grease applied.

LV Electrode

Not required where the HV and LV earths are to be bonded, as the HV earth will fulfil the necessary requirements.

If separate earths are required (See section 3.20.7.3.) a minimum distance of 9m must be maintained between this electrode and any 11kV electrode, the substation compound and any other metalwork connected to the HV earth. The LV electrode will be connected to the LV neutral busbar in the LV cabinet via at least 9m of 70mm² PVC covered copper conductor. The PVC conductor must be compression bonded to the LV earth electrode at one end, and to the neutral bar of the distribution cabinet at the other, via CE approved connectors.

High Voltage Cables

The outer metal sheath of 11kV cables will be bonded to the switchgear bonding bar via a flexible strip or braid having a current carrying capacity equivalent to at least 50mm² copper. This is to be of sufficient length to accommodate a fault current indicator.

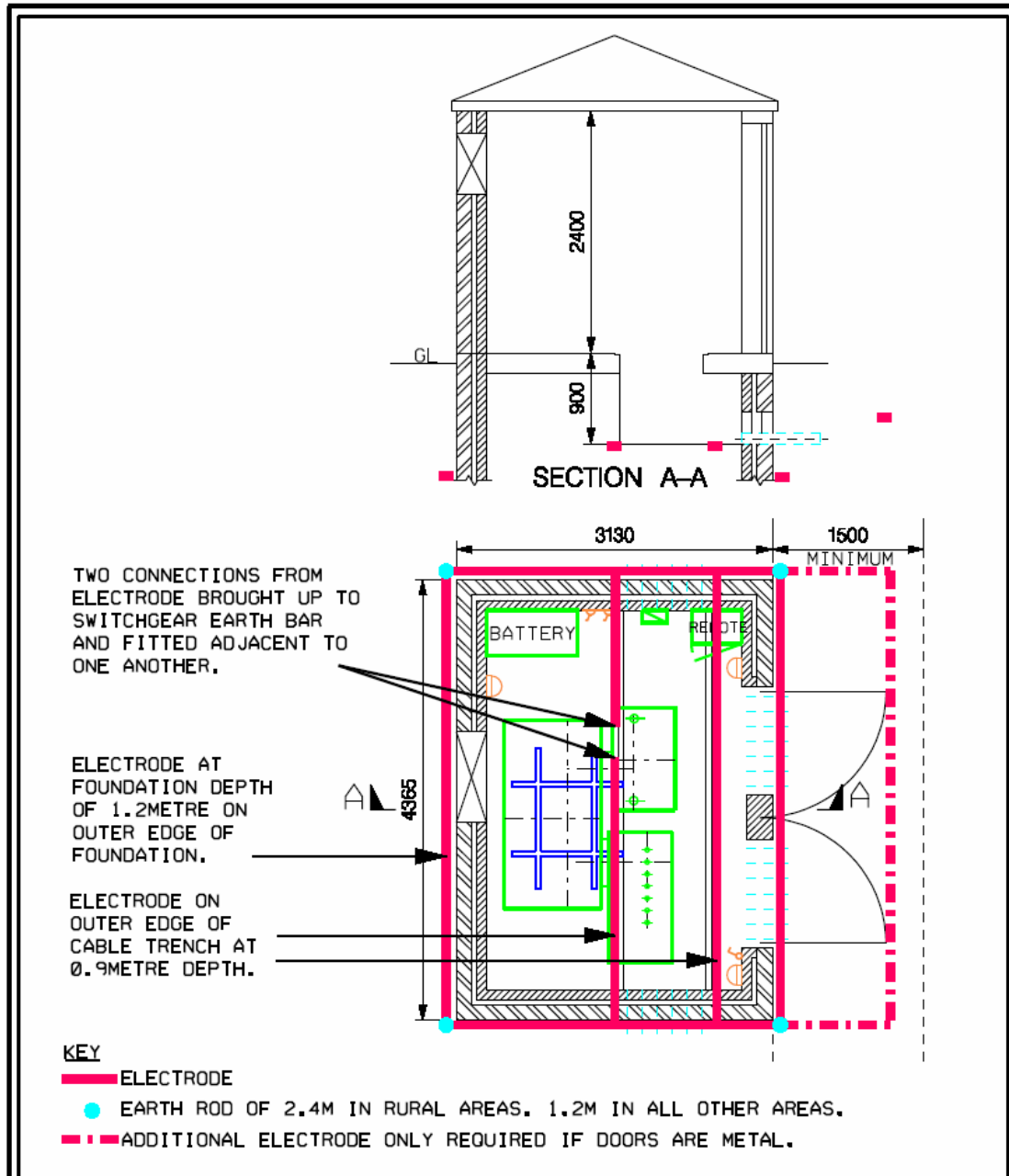
Low Voltage Cables

For modern CNE cables, the outer sheath of the cable will be connected to the neutral bar in the distribution pillar using a CE approved compression or mechanical connector.

For SNE cables the outer sheath and armouring should be bonded together and connected to the earth bar in the distribution pillar. The neutral conductor is then connected to the neutral bar.

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 45 of 87

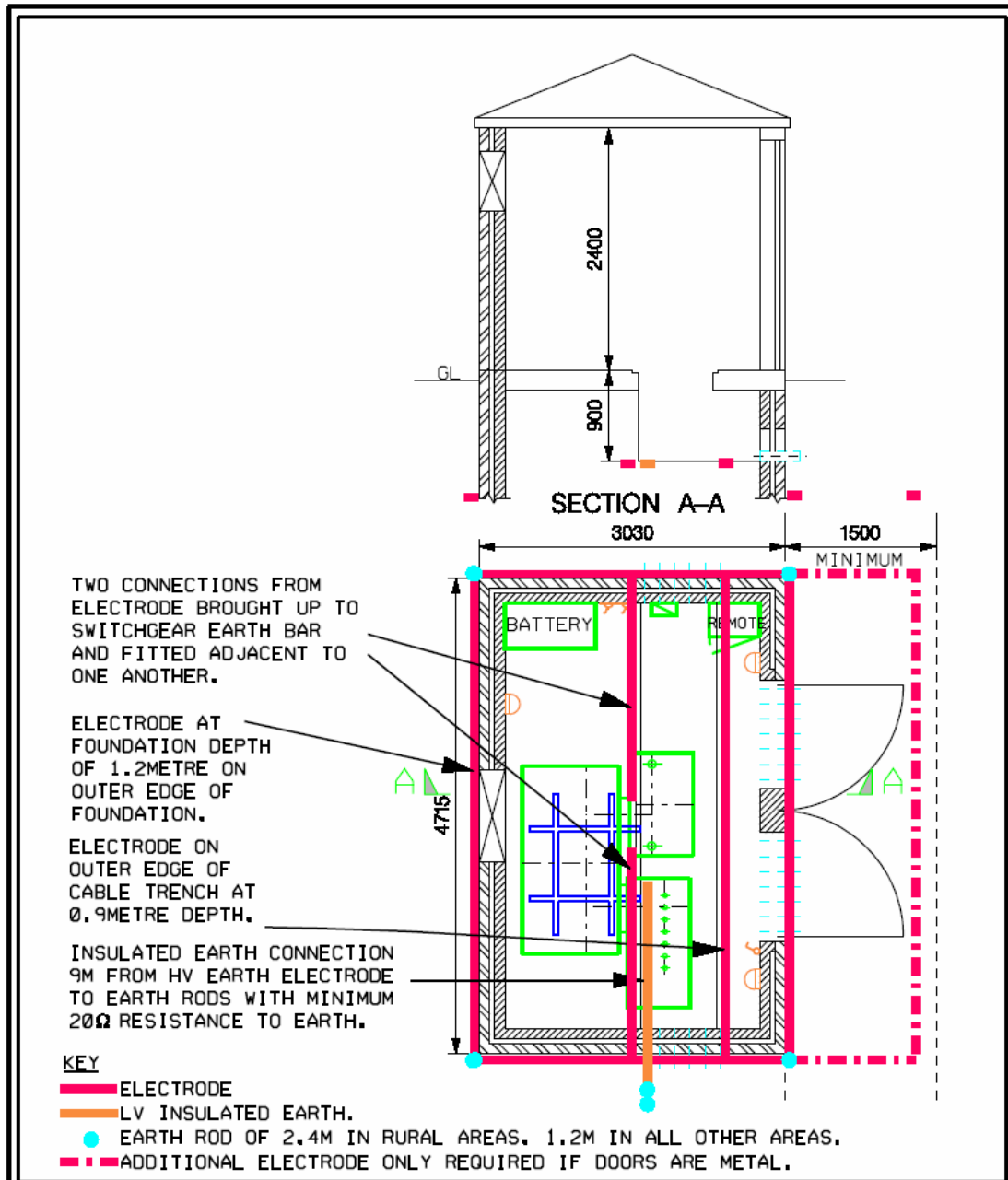
Arrangement 1 - Ground Mounted Substation with close coupled switchgear and Combined HV/LV Earth



CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF		
Manufacturer Details		EARTHING ARRANGEMENT AT GROUND MOUNTED SUBSTATIONS GROUND MOUNTED SUBSTATION WITH CLOSE COUPLED SWITCHGEAR WITH COMBINED HV AND LV EARTH		
Sheet No.	Scale	Document Details		
	1:50	CONSTRUCTION STANDARDS		
Prepared By	Brian James	Ref No.	C944571	Historic Drawing No.
Revised	03/05/2007	Grid Reference		Y001S10
Date Issued	31/01/2007	Checked By	J. PAINE.	Revision
				00

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 46 of 87

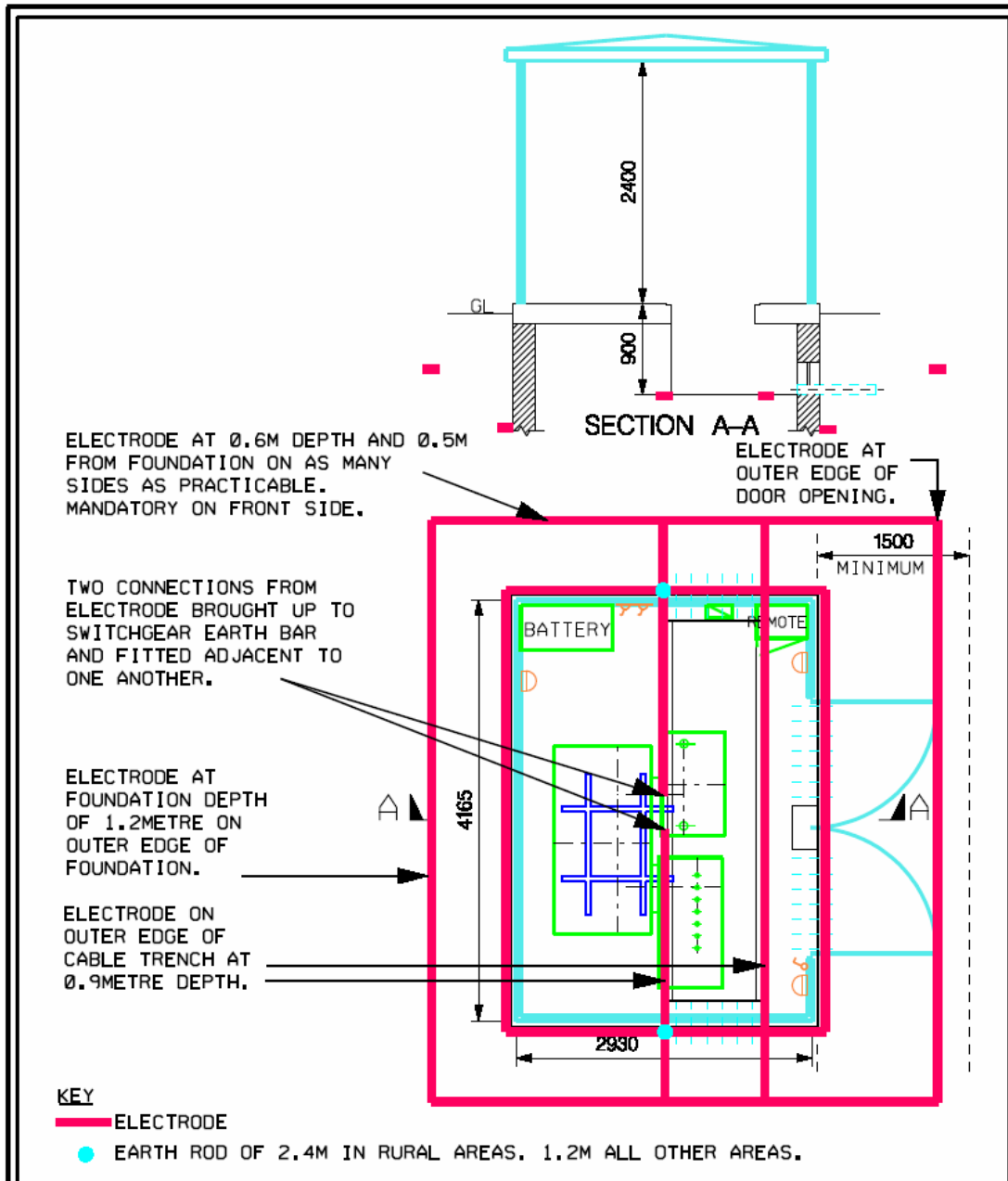
Arrangement 2 - Ground Mounted Substation with close coupled switchgear and separate HV/LV Earth



CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		EARTHING ARRANGEMENT AT GROUND MOUNTED SUBSTATIONS UDE GROUND MOUNTED SUBSTATION WITH CLOSE COUPLED SWITCHGEAR AND SEPERATE HV/LV EARTH	
Sheet No.	Scale	Document Details	
	1:50	CONSTRUCTION STANDARDS	
Prepared By	Brian James	Ref No.	C951997
Revised	12/09/2007	Grid Reference	Y001S10
Date Issued	12/09/2007	Checked By	J. PAINE.
		Revision	00
		Notes	

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 47 of 87

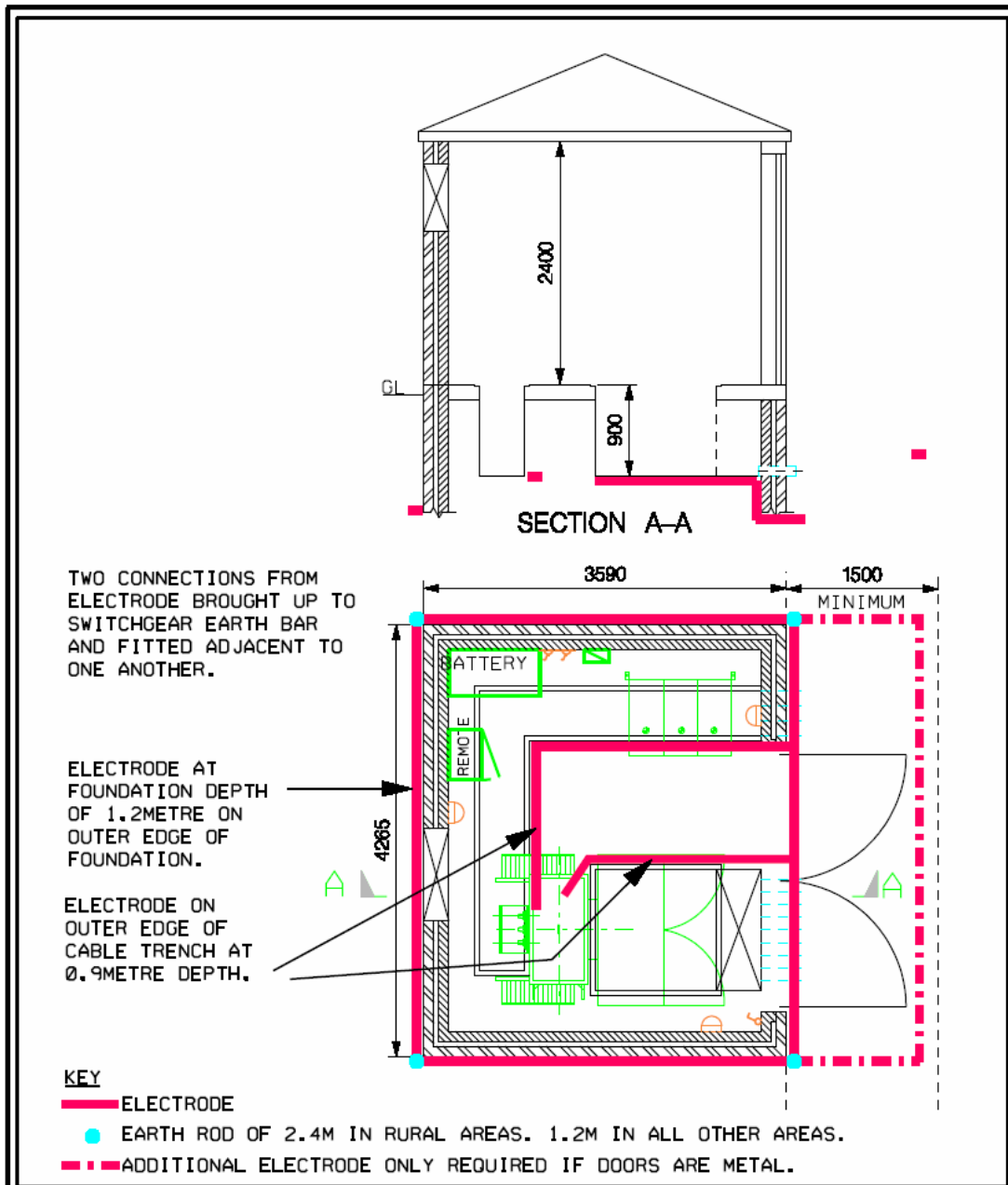
Arrangement 3 - Steel Kiosk Type Ground Mounted Substation with Combined HV/LV Earth



CE Electric UK		Lloyds Court, 78 Gray Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		EARTHING ARRANGEMENT AT GROUND MOUNTED SUBSTATIONS STEEL KIOSK TYPE GROUND MOUNTED SUBSTATION WITH COMBINED HV/LV EARTH	
Sheet No.	Scale	1:50	
Prepared By		Document Details	
Brian James		CONSTRUCTION STANDARDS	
Revised	Grid Reference	Ref No.	Historic Drawing No.
31/01/2007		C944570	Y001S10
Date Issued	Checked By	Revision	Notes
31/01/2007	J. PAINE.	00	

Document Reference	IMP/010/011		
Version	1	Date of Issue	December 2007
		Page	48 of 87

Arrangement 4 - Ground Mounted Substation with separate HV switchgear and Combined HV/LV Earth



CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		EARTHING ARRANGEMENT AT GROUND MOUNTED SUBSTATIONS	
Sheet No.	Scale	GROUND MOUNTED 20KV SUBSTATION	
	1:50	WITH SEPERATE HV SWITCHGEAR	
Prepared By	Document Details	WITH COMBINED HV AND LV EARTH	
Brian James	CONSTRUCTION STANDARDS		
Revised	Grid Reference	Ref No.	Historic Drawing No.
31/01/2007		C944575	Y001S10
Date Issued	Checked By	Revision	Notes
31/01/2007	J. PAINE.	00	

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	49 of 87

3.20.8 Pole-Mounted Substations

The cost of installing earth electrodes at an isolated pole-mounted substation and associated LV network to achieve a resistance to earth value capable of restricting the EPR to less than 430V under HV fault conditions will normally be excessive.

Therefore, the HV earth electrode at the pole-mounted substation shall always be separated from the LV earth electrode.

The HV earth electrode and the LV earth electrode shall each have a resistance to earth less than 20 ohms; the value will be measured by an approved method.

A minimum distance of 9 metres will be maintained between any uninsulated parts of the HV and LV earth electrodes which are below ground level. The required separation will normally be obtained by earthing the LV neutral conductor one span away from the substation, Drawing C953076 Fig A. (over page) Connections to earth electrodes will be insulated from earth to a depth of at least 600mm below ground level using the conductors specified in Appendix E.

Consideration must be given to maintaining HV/LV earth physical segregation including taking into account the presence of metallic sheathed cables.

If a metallic sheath LV cable is connected at the substation its metallic sheath must be connected to the neutral conductor. Separation of the electrodes in this case may be achieved by connecting the transformer tank to a remote earth electrode by a green coloured insulated lead as shown in Drawing C953076 Fig B. The HV earth electrode lead on the pole will be spaced at least 75mm from the LV earth-lead and the sheath of the LV cable.

If, at an existing pole-mounted substation, a common HV and LV earth electrode is installed and its resistance to earth is greater than 1 ohm then the HV earth shall be separated from the LV earth. The existing earth electrode shall normally be used for the LV earth and a new electrode shall be installed for the HV earth.

Pole mounted substations supplying mobile phone base stations on transmission towers. shall be designed and installed in accordance with DSS/022/001 (Code of Practice for Low Voltage Connections to Mobile Telephone Base Stations with Antennae on High Voltage Structures)

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 50 of 87

FIG A

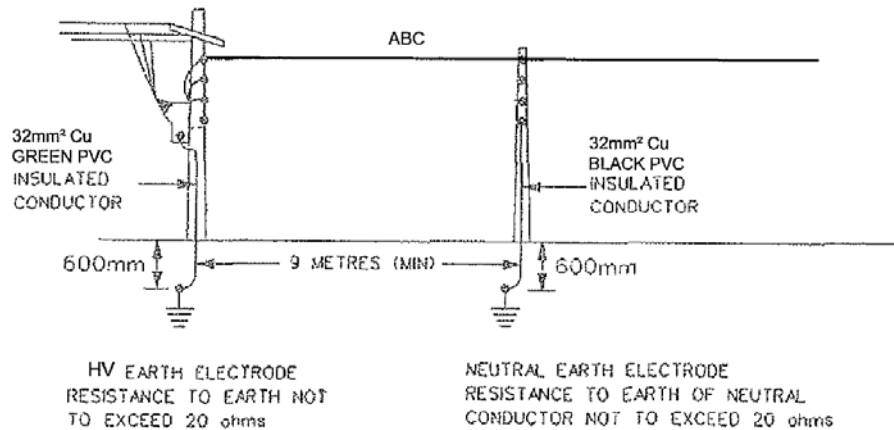
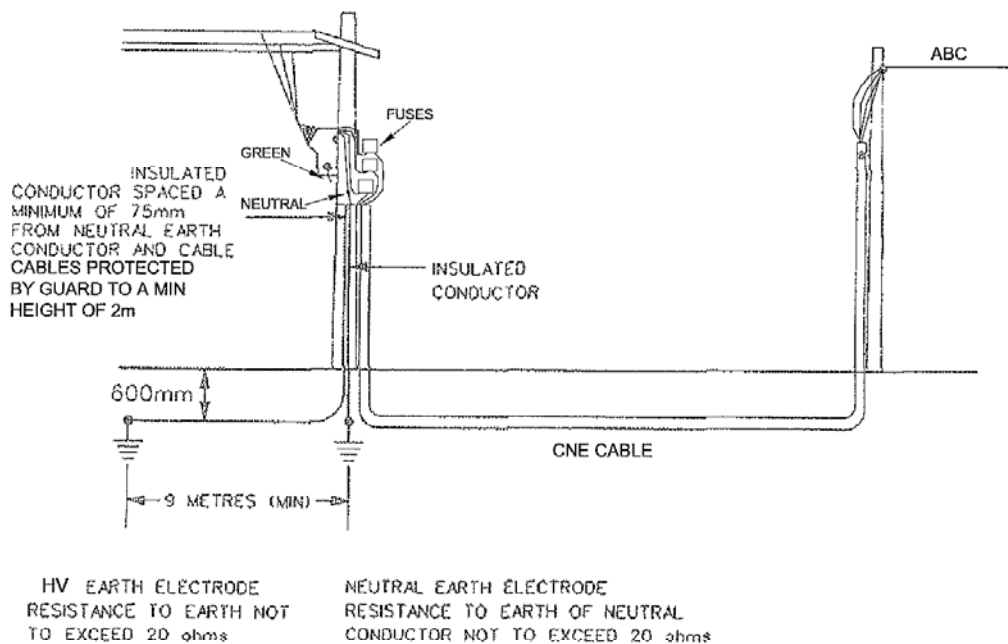


FIG B



CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF		
Manufacturer Details		EARTHING ARRANGEMENTS AT POLE MOUNTED SUBSTATIONS WITH SEPERATE HV AND NEUTRAL EARTH ELECTRODES		
Sheet No.	Scale N.T.S.	Type PLANT DIAGRAMS GENERAL		
Prepared By Barbara Gordon	Grid Reference	Ref No.	C953076	Historic Drg.No. C953076
Revised 22/10/07	Checked By J PAINE	Revision	Notes	
Date Issued 17/10/07				

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	51 of 87

4. References

4.1 Internal

DSS/01/0016 Guidance on Engineering Recommendation G39 (Earthing Arrangements for Street Lighting Furniture)

DSS/022/001 – Code of Practice for Low Voltage Connections to Mobile Telephone Base Stations with Antennae on High Voltage Structures

NSP/002/001: Specification for Earthing Materials

NSP/004/041: Specification for LV ABC Overhead Lines

NSP/004/043: Specification for Overhead Services, Surface Wiring and Eaves Wall Mains

4.2 External

BS 951: Electrical Earthing. Clamps for Earthing & Bonding. Specification

BS 62035 - Code of practice for protection against lightning

BS 7361 Cathodic Protection.

BS 7375: Code of practice for Distribution of electricity on construction and building sites

BS7671 - Requirements for Electrical Installations (IEE Wiring Regulations) - Current Edition

Engineering Recommendation G59/1- Recommendations for the Connection of Embedded Generating Plant to the Public Electricity Suppliers Distribution Systems (Amendment 1 (1992) & Amendment 2 (1995))

Engineering Recommendation G78 - Recommendations for the provision of LV Connections to Mobile Telephone Base Stations with Antennae on High Voltage Structures.

Engineering Recommendation G83/1 - The connection of small-scale embedded generators (Up to 16A per phase) in parallel with public low voltage distribution networks

Engineering Recommendation G87 Guidance for the Provision of Low Voltage Supplies to Multi Occupancy Buildings

Engineering Recommendation G12/3 - National Code of Practice on the Application of Protective Multiple Earthing

Engineering Recommendation G14 - Protective Multiple Earthing; Recommended Principles for Testing to ensure Correct Polarity

Engineering Recommendation G39/1 - Model Code of Practice covering electrical safety in the planning, installation, commissioning and maintenance of public lighting and other street furniture

Engineering Recommendation P04/1: Services to British Telecom public telephone housings

Engineering Recommendation P24, 1984: AC traction supplies to British Rail (Addendum No.1 1990)

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	52 of 87

ENA Technical Specification 41-24: Guidelines for the design, installation, testing and maintenance of earthing systems in substations

ENA Technical Specification 43-94: Earth rods and their connectors

Guidance Note GS 24 - Electricity on Construction Sites

Guidance Note HS(G)41 - Petrol Filling Stations: Construction and Operation

The Electricity Safety, Quality and Continuity Regulations 2002.

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 53 of 87

5 Definitions

"Aerial Bunched Conductor (ABC)" refers to a single bunched LV overhead cable construction

"Bonding Lead" means a conductor by which items of metalwork are connected together electrically, not normally for the purpose of carrying current, but so as to ensure a common potential.

"Branch Lines" means any electric line through which energy may be supplied to 5 or more customers from any distributing main

"Business Owner" means the manager (or his representative) within the business responsible for the business process

"Combined Neutral and Earth (CNE)" refers to a network where the earth return and neutral return are provided by the same conductor

"Customer" means any body or person supplied or entitled to be supplied with energy by CE and will include those customers on un-metered supplies.

"Customer's Installation" means the electrical apparatus under the control of the customer on the customer's premises together with the wiring connecting this apparatus to the supply terminals. A meter and cutout owned by CE shall not form part of the customer's installation.

"Customer's Premises" means any area or building in the sole occupation of the customer.

"Distributing Main" means the portion of any main which is used for giving supply to branch lines, service lines and multi-service lines for the purpose of general supply.

"Distributor" means a person who owns or operates a network, except for a network situated entirely offshore or where that person is an operator of a network within the meaning of Part I of the Railways Act 1993

"Earth-Continuity Conductor" means a conductor including any clamp connecting a part of an installation or other accessible metalwork which is required to be earthed, to another such part or to the supply neutral conductor.

"Earth Electrode" means a metal rod, plate or strip conductor buried in the earth for the purpose of providing a connection with the general mass of earth.

"Earthing Lead" means the final conductor by which the connection to the earth electrode is made.

"Earthing Terminal" means the terminal provided by CE and directly connected to the supply neutral conductor at the supply terminals.

"Earth Potential Rise (EPR)" means the rise in potential from a zero earth reference value during an earth fault on an adjacent HV or EHV circuit.

"EHV" Means 33kV, 66kV or 132kV

"Global Earth" means an equivalent earthing system created by the interconnection of local earthing systems made up of earth electrodes and cables with metallic sheaths in direct contact with earth that ensures, by the proximity of the earthing systems, that there are no dangerous touch voltages

"HV" High Voltage means above 5.5kV up to and including 20.5 kV

"Multi-service Line" means any electric line through which energy may be supplied to two, three or four adjacent customers from any distribution main or branch line or substation.

Document Reference		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	54 of 87

"Protective Multiple Earthing (PME)" - the technique of using the supply neutral conductor of a LV distribution system for earthing of services to customers and street lights or other street furniture.

"Protective Neutral Bonding (PNB)" refers to the situation where there is only one point in a network at which consumer's installations are connected to a single source of voltage, in such a case the supply neutral conductor connection to earth may be made at that point, or at another point nearer to the source of voltage.

"Residual Current Device (RCD)" means a mechanical switching device or association of devices intended to cause the opening of the contacts when the residual current attains a given value under specified conditions

"Service Lines" means any electric line through which energy may be supplied to a customer from any distributing main on branch line or directly from a substation.

"Separate Neutral and Earth (SNE)" refers to a network where the earth return and neutral return are provided by separate conductors.

"TN-C" means a systems where the neutral and protective functions are combined in a single conductor throughout the system. Earthed concentric wiring or Earthed Sheath Return Wiring are both examples of a TN-C system also known as CNE systems.

"TN-S" means a systems where there are separate neutral and protective conductors throughout the system. A system where the metallic path between the installation and the source of energy is the sheath and armouring of the supply cable is a TN-S system also known as SNE systems.

"TN-C-S" means a systems where the neutral and protective functions are combined in a single conductor but only in a part of a system. Systems with PME applied to them are examples of a TN-C-S system, where the distribution system between the source of energy and the installation is TN-C and the installation itself is TN-S. PNB is also a variant of a TN-C-S system also known as a mixed system.

"TT" means a system that has one or more points of the source of energy directly earthed and the exposed and extraneous-conductive parts of the installation are connected to a local earth electrode or electrodes that are electrically independent of the source earth(s). An example would be where the customer provides his own independent local earth and provides earth fault protection by use of a RCD

"Where Practical" in the context of this document means where it is physically and technically possible to achieve within the budgetary context of the work programme or capital scheme that the activity is taking place under.

Document Reference		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 55 of 87

6 Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Jim Paine	Protection and Technical Services Manager	Jim Paine	28/11/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Andy Ellam	Network Performance & Risk Manager	Andy Ellam	28/11/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Dave Miller	Network Investment Engineer	Dave Miller	28/11/07
Ged Hammel	Overhead Line Standards Manager	Ged Hammel	28/11/07
Glen Hodges	Programmes and Strategic Design Leader	Glen Hodges	28/11/07

6.4 CDS Assurance

I sign to confirm that this document has been assured for issue on to the DBD system.

		Sign	Date
Sean Johnson	CDS Administrator	Sean Johnson	14/12/07

6.5 Approval

Approval is given for the content of this document.

		Sign	Date
Mark Nicholson	Head of System Strategy	Mark Nicholson	29/11/07

6.6 Authorisation

Authorisation is granted for publication of this document.

		Sign	Date
Mark Drye	Director of Asset Management	Mark Drye	29/11/07

Appendix A Customer Installation to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	56 of 87

Customers' Installations

Customers installations can only be connected to a PME terminal if the bonding conforms to BS7671 (IEE Wiring Regulations)

1. Bonding and Earthing

The customer's water, gas and any other metallic services (except metalwork forming part of a telegraphic, telephone or signalling circuit) shall be bonded to the CE earthing terminal block either separately or by means of a common bonding lead having a minimum cross sectional area as specified in Appendix E. Conductor shall be stranded copper, green/yellow PVC insulated cable. Any other metal structures which might come into electrical contact with earth and which are in easy reach of any exposed non-current-carrying metalwork of the customer's installation shall also be bonded to the earthing terminal.

Where doubts arise on the need for bonding particularly items of extraneous metalwork the criteria to follow are:-

(a) All structural metalwork which is embedded in the earth and is situated such that a person may make simultaneous contact with it and any other exposed metalwork which is to be connected to the earthing terminal must be bonded by a main equipotential bonding lead having a minimum cross-sectional area as specified in Appendix E.

(b) Supplementary local bonding connections may sometimes be required to maintain the equipotential zone but exposed extraneous fixed metalwork within the zone which is not in contact with earth need not be bonded unless it is likely to come into fortuitous contact with other earthed or bonded metalwork. Bonding connections with cross-sectional areas in accordance with the current edition of the IEE Wiring Regulations are then required.

Connections to the service pipes shall be by means of clamps suitable for the pipe diameters which will avoid electrolytic action. These should be made as near as possible, with the exception of the gas service, to the point of entry of the service into the customer's premises.

In the case of gas services, the connection shall be made as near as possible to the outlet union on the customer's side of the gas meter, and before any branch pipe, and in any case no further than 600 mm from the meter. If, for actual or possible cathodic protection, a gas distribution company has installed an insulated insert between the incoming service pipe and the gas meter, the incoming service pipe shall not be bonded. Incoming unused gas service pipes should not be bonded but, if such pipes extended well inside the building, the pipe should be insulated in such a way that the insulation cannot easily be removed.

If the electricity service cable is mechanically protected, then any protecting metalwork shall be connected to the earthing terminal block.

The bonding of the customer's water, gas and any other metallic services with the exception of any CE equipment is the responsibility of the customer.

2. Arrangement at Customer's Intake Unit

Appendix A Customer Installation to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	57 of 87

Typical customer arrangements are shown in drawing C953036

The earth terminal made available to the customer will be by means of an earth terminal block securely bonded to the neutral conductor. This is achieved in the latest design of unit by bonding the neutral external connection to the CE earth terminal block, where provided, or the customer's earth terminal block and, on the older type, by bonding the neutral internal link on the incoming or outgoing side (as convenient) to the CE earth terminal block. The bonding lead shall be 16 mm² minimum green/yellow PVC insulated cable or half the size of the neutral meter tail whichever is the larger. Any metallic sheath or armouring of underground service cables shall be bonded to the earth terminal block or neutral terminal as appropriate.

On temporary supplies and in other situations where we are unable to verify that the requirements of bonding and earthing as described in Section 1 above have been complied with, the earthing terminal block shall be rendered inaccessible so as to prevent unauthorised connection.

The size of common bonding lead between the customer's metallic services and the CE earthing terminal block will be a minimum of 10 mm² and that between the neutral link and the CE earthing terminal block will be a minimum of 16 mm². Where the expected loads of a customer's installation require it, the size of these connections shall be increased in accordance with the appropriate values shown in Appendix E.

3. Installation of Earthing Clamps at Customer's intake unit on existing underground and overhead networks

In existing customer's installations where there is an earth return path to the substation via the underground cable sheath or separate earth conductor then, when the opportunity arises as detailed in Section 3.3.2, earthing clamps of an approved pattern shall be fixed, one to the cable sheath and one to any armouring. The clamps shall be bonded together and connected to a CE earthing terminal block on the meter board, to which the customer's earth continuity conductor may be connected. If the service cable contains a separate earth conductor, this shall be connected to the earthing terminal block.

The customer shall be informed by a standard letter that the earth terminal has been provided and is available for his use.

It should be noted that private contractors' staff are not permitted to fix the clamps under any circumstances.

4. Warning Notice

A warning notice shall be fixed at the customer's service position indicating that the earth terminal is PME.

When a new PNB earth terminals is installed the appropriate label will be fixed at the service position. If a supply to a single customer employing PNB is later extended to become a PME network feeding a number of customers, then the PNB warning notice at the original customer's premises shall be replaced by the PME type.

5. Customer's Service Polarity Testing

Appendix A Customer Installation to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	58 of 87

On a PME system the connection of the live conductor of a service to the neutral connector block on the customer's premises creates a dangerous situation where all the customer's metalwork bonded to the CE earth terminal block becomes live at line voltage. Under these circumstances, the earth in the vicinity of the premises also tends to rise to line voltage, and tests intended to establish polarity may be misleading and may indicate correct polarity even though it is reversed.

Polarity tests will be carried out on all new customer installations or when any work has been carried out which could have affected the customer's polarity.

Appendix B Procedure for Customers with inadequate earths to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	59 of 87

Procedure to be applied where a Customer is reported to have inadequate earthing

The following procedure is to be applied where it is reported that an existing CE customer has either no effective means of earthing their installation or if the earth terminal provided is inadequate.

Where CE Electric UK have not previously provided an earth terminal to the customer or where the customer has been reliant on the water pipes for earthing

When a representative from CE Electric UK identifies through testing or is made aware that a domestic customer does not have an earth or has an inadequate earth they will advise the customer that their installation is not up to current standards and could cause danger in certain situations (e.g. a fault on an electrical appliance within the property). The customer will be informed to contact a qualified electrician to make their electrical installation safe and to confirm when this work has been completed.

This information will be presented in the form of a standard letter, which will be completed on site and presented to the customer. The customer should be advised that if they do not employ an electrician to rectify the defective earthing within 3 weeks (15 working days) the supply to the premises will be disconnected on safety grounds under regulation 26 of the Electricity Safety, Quality and Continuity Regulations 2002 [1].

The Company representative will leave the standard sample letter 1 below with the customer and inform the business owner of this action. The business owner will record the fact that a notification letter has been left with the customer. The business owner will send an e-mail to the Services Programme Manager giving details of the customer and the date that the notification letter was issued [2].

If the Customer has not contacted business owner confirming that the work has been completed within 28 days of the notice being issued the business owner will write to the customer with a confirmation letter sample letter 2 below [3].

If the customer does not confirm that the remedial work has been completed within 7 days of issuing the 2nd letter a the business owner will arrange for a business representative to check the customers earth and to remove the cut-out fuse if it is established that the remedial work has not been completed [4] and leave a confirmation letter sample letter 4 below.

Where CE Electric UK have provided an earth terminal to a customer and the value of earth loop impedance as tested from the customers cut-out is high

- 1) Where CE Electric UK have provided an earth terminal to a customer and the value of earth loop impedance as tested from the customers cut-out is high (above 0.35 ohm for CNE supply and above 0.8 ohm for an SNE supply) [5] the high earth loop impedance should be dealt with as a fault repair to the network earth. The Asset Management Department will be informed by e-mail of every such instance.
- 2) Where a representative from CE Electric UK identifies through testing that the high earth loop impedance is on the customers premises the customer will be advised that their earthing installation is not to current standards and that they need to contact their own

Appendix B Procedure for Customers with inadequate earths to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	60 of 87

electrician to rectify the situation the letter in sample letter 3 below will be left with the customer.

Notes:

Note [1] With reference to Regulation 26 of the ESQC Regs

26(1) permits the DNO to issue a notice requiring the customer to rectify defects within 10 days (the guidance notes cite an example where there are live conductors sticking out from a consumer unit).

26(2) If the work hasn't been done within 10 days (for the example in the guidance notes) we can disconnect - and leave another letter.

26(3) We can disconnect without giving notice on the grounds of safety. It's difficult so see how an inadequate earth could be more dangerous than live wires sticking out of a consumer unit, and hence it probably would not be appropriate for us to disconnect immediately on the grounds of safety assuming the earth has been defective for some time. So it would seem reasonable to give notice of disconnection and provide a reasonable time for the defects to be rectified.

26(4) We can reconnect as soon as the work's been done.

10 working days is discussed in the guidance notes for live exposed conductors, a no earth situation will have a lower probability of causing harm as it needs another event such as a faulty appliance to cause harm. 21 days is suggested as a reasonable time for customer to make arrangements with an electrician without unduly delaying correction of the defects.

Note [2]

The Earthing Policy Manager will keep a record of the occurrences where an earth is not available from the network or where the network earth is inadequate to determine if they are isolated incidents or if there are areas of the network which need remedial attention.

The Services Programme Manager needs advance warning so that he can inform his contractors that there may be an urgent request for an earth terminal from the customers electrician. The Services Programme Manager will inform the Earthing Policy Manager of any request for an earth, the remedial action taken and the state of the network.

There are 3 likely scenarios:

1. The customer contacts a qualified electrician whom makes a request for an earth terminal on behalf of the customer. In this case if an earth can be provided for the customer with work on the service or cut-out position it will be provided via scheduled work.. If the provision of a network earth needs substantial work on the LV main cable to provide, the electrician will be advised that an earth cannot be provided from the network and advised to make his own earth arrangements. In this case the Services Programme Manager will inform the Earthing Policy Manager of the work required to provide an earth and the fact that the customer has been informed to provide their own earth. The defective network will be brought up to specification through a work programme.
2. The customers electrician provides a customer earth and corrects the defects without requesting a Network earth.
3. The customer does not take any action to remedy the defects.

Note [3]

Appendix B Procedure for Customers with inadequate earths to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	61 of 87

The business owner will not send letter 2 until 28 days after the notification date to allow for a letter to be received from the customer if the work was completed on the 21st Day.

Note [4]

The company may be criticised if customers have their supplies disconnected, however if there is an incident at the property and a person is electrocuted, or there is a fire resulting directly from the defective earth installation that the company was aware of the company would have failed in its duties under the ESQC Regs. However the customers least likely to respond to the notification are the old or vulnerable in society, the policy may need to be adjusted to provide some form of support for these customers. The company can not take any responsibility for work on the customers network.

Note [5]

If there is a high earth loop impedance on the network it is an indication that there is a fault in the earth system. Where the system is CNE this could result in an open circuit neutral/earth resulting in wide spread damage. Such faults should be dealt with by repairs with the appropriate level of priority placed on the actual fault condition. It is important that the Earthing Policy Manager is informed of every occurrence of earth or neutral/earth fault on the network.

Appendix B Procedure for Customers with inadequate earths to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	62 of 87

Sample Letter 1 – on company letter headed paper

To the occupier of:

Add contact address Business Owner

Dated

To the occupier,

A representative from CE attended your property on the date of this letter and established that the earth system for your electrical installation was defective.

These defects are listed below:

The safe operation of your electrical installation and portable appliances rely on an effective earth, the defects identified could give rise to danger.

Under the Electricity Safety Quality and Continuity Regulations 2002 a Distribution Network Operator cannot knowingly supply electricity to a defective installation. Regulation 26 gives the authority to disconnect defective installations that could give rise to danger.

This letter gives notice of disconnection for reasons of safety. .

What you must do now.

You must contact a qualified electrician and have them rectify the defects to your electrical installation listed in this letter, within 21 days of the date of this letter.

When the work is completed inform the **** in writing.

If the defects listed in this letter are not rectified within 21 days of the date of this letter. Your supply will be disconnected for reasons of safety

Signed on behalf of YEDL/NEDL

Appendix B Procedure for Customers with inadequate earths to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	63 of 87

Sample Letter 2 – on company letter headed paper

To the occupier of:

Add contact address Business Owner

Dated

To the occupier,

A representative from CE attended your property on the ***** established that the earth system for your electrical installation was defective.

The defects identified were

You were also left notification that electrical supply to your property would be disconnected after 21days if the defects, had not been rectified by a qualified electrician.

Please confirm by return of post that the defects have now been rectified.

If you do not confirm that the defects have been rectified or make contact by phone or letter to the contact point above the electricity supply to your property will be disconnected for reasons of safety.

Signed on behalf of YEDL/NEDL

Appendix B Procedure for Customers with inadequate earths to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	64 of 87

Sample letter 3 – on company letter headed paper

To the occupier of:

Add contact address Business Owner

Dated

To the occupier,

A representative from CE attended your property on the date of this letter and established that the earth system for your electrical installation was defective.

These defects are listed below:

The defects identified could give rise to danger in certain situations such as a fault on an electrical appliance within your property.

What you must do now.

You must contact a qualified electrician and have them rectify the defects to your electrical installation listed in this letter.

Signed on behalf of YEDL/NEDL

Appendix B Procedure for Customers with inadequate earths to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	65 of 87

Sample letter 4 – on company letter headed paper

To the occupier of:

Add contact address Business Owner

Dated

To the occupier,

A representative from CE attended your property on the date of this letter to disconnect your electrical supply because you have failed to carry out remedial work on the earth of your electrical installation.

You were given a notice of disconnection on ***** which included details of the defects.

The disconnection has been carried out on safety grounds under authority of The Electricity Safety Quality and Continuity Regulations 2002, Regulation 26.

Your property will not be reconnected until you have confirmed in writing that the defects have been corrected.

Signed on behalf of YEDL/NEDL

Appendix C Guidance on the Control of EMC Problems to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	66 of 87

General Guidance for Minimising EMC Problems.

This is becoming an important issue due to the proliferation of computers and other IT type business and entertainment equipment within commercial, industrial and domestic premises. Systems should be able to operate without interfering with each other.

Earthing has an important role in EMC, but this must never be allowed to compromise its primary safety role.

The following general principles should be followed to help reduce the impact of EMC problems on an installation:-

Have a combined earthing/electrode system with multiple ground connection points. This will form part of the common bonding system, which will include extraneous metalwork, steel beams, rebar, pipes etc. Wherever possible the company earthing terminal would form part of this. Internal equipment, lightning protection etc., would then be connected to the common bonding system in a manner required by the customers building services engineer.

Wherever possible, the internal power supply should be of the TN-S type as this has the best EMC performance. Avoid providing multiple PME supplies to single buildings particularly where there are paths for neutral current diversion through structural steelwork, bonding etc. Although hard to manage at large, multi-occupancy buildings, this is essential to prevent significant EMC issues.

Incoming and outgoing cables are best grouped at one point, near which the main incoming earthing terminal and connections to a closely located earth electrode exist. The earth sheaths and screens of the incoming power supply cable should be connected directly to the main earth terminal or (if the earths are not to be connected) have a surge protection device fitted.

Inter-building cables should preferably be in shallow metal troughs, which are multiply earthed and bonded to an integrated electrode system at either end.

The following is a list of other precautionary measures, which may be applied to minimise EMC problems.

Separate control cable routes from those of power cables

Install cables in trefoil rather than flat formation.

Avoid secondary cable runs in parallel with busbars or power cables.

Route control cables to avoid single-phase transformers and inductances.

Avoid cable earth loops.

Ensure that all wires associated with the same circuit are contained in one cable or grouped along one route.

Auxiliary cable routes to have radial rather than ring configuration.

Use screened, twisted pair cables for data transfer.

Suitable shielding of secondary circuit cables.

Group circuits associated with the same function, wherever possible.

Any equipment used should have the appropriate immunity rating and should be selected and grouped according to its working environment. Filters and voltage limiting devices should be used where necessary.

Appendix D Earth Electrode Testing to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	67 of 87

Earth Electrode Testing

1. Method of Checking Adequate Electrical Separation between Two Electrodes

Three tests are carried out (see Drawing C953068). In the first test, the resistance of Electrode A to the general mass of earth is measured by an earth megger or the equivalent, according to the instructions issued with the instrument; let this resistance be R_{ae} . In the second test, the same measurement is carried out on Electrode B; let the result be R_{be} . In the third test, the resistance between A and B is measured let this be R_{ab} .

If the resistance areas of the two electrodes are adequately separated then R_{ab} must be equal to or greater than $0.9 (R_{ae} + R_{be})$.

It is important when measuring R_{ae} and R_{be} to ensure that the voltage and current leads are run out to adequate distances.

2. Method of Measuring the Overall Resistance to Earth of a Multiple Earthed Neutral Conductor or Other Extensive Earthing Systems

The combined resistance to earth of an extensive earthing system provided by neutral electrodes cannot be conveniently measured. Therefore the resistance to earth of each earth electrode shall be measured and the combined resistance to earth shall be calculated by assuming that the individual resistance values can be regarded as connected in parallel without their resistance areas overlapping.

The combined resistance to earth is given by:

$$\frac{1}{R_c} = \frac{1}{R_{ae}} + \frac{1}{R_{be}} + \frac{1}{R_{ce}} + \dots$$

where R_c = Combined resistance to earth of the individual electrodes

R_{ae}
 R_{be}
 R_{ce} = Resistance to earth of individual electrodes.

It is important that when measuring each electrode resistance to earth that voltage and current leads are run out to adequate distances to ensure that there is no overlap between the resistance.

3. Measurement of Resistance of Individual Earth Components using a clip on earth tester

At a distribution substation, the following items are those that, in parallel, provide the earth resistance value.

- (i) 20/11/6.6kV electrode at the substation.
- (ii) LV electrode just beyond the substation.
- (iii) 20/11/6.6kV cable sheaths.
- (iv) LV earths beyond substation.

An individual measurement of each of the first three can be made by connecting a clip on, (contact-less) earth tester around the earth connection between it and the switchgear earth bar. The individual values obtained should be noted and compared to the overall value obtained by

Appendix D Earth Electrode Testing to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	68 of 87

measurement. The clip on meter measures the value of the electrode under test in series with the rest of the connected electrodes. The measured value will only be accurate if the other electrodes in parallel have a low value compared to the one under test. Figures D.1 and D.2 show the arrangement for a pole type substation and the electrical equivalent circuit.

Figure D.13: Current Flows for a Loop Resistance Measurement of the LV Earth

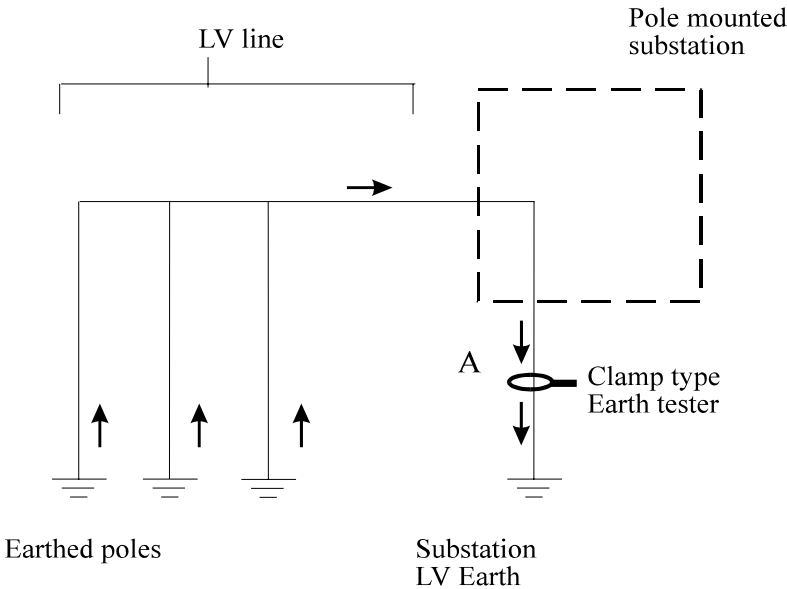
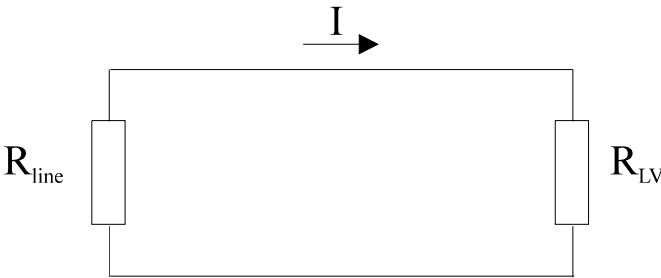


Figure 1.2.4: Equivalent Circuit of a Loop Resistance Measurement



R_{line} is the resistance of the LV overhead earth wire and pole earths, in parallel.
 R_{LV} is the earth resistance of the substation LV earth.

The resistance measured by the clamp-on earth tester (R_{total}) will be the sum of the resistance in the loop, i.e. ($R_{LV} + R_{line}$). It is assumed that the resistance of the return path (R_{line}) is small enough to be considered negligible against the larger R_{LV} value, so that R_{total} is approximately equal to R_{LV} . If a relatively low resistance return path is not available, then this type of measurement cannot be made accurately.

Appendix E Type and Size of Earth Connections to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	69 of 87

Type and Size of Earth Connections

The types and size of earthing and bonding connections shall be as detailed in the tables below

LOCATION	EARTHING AND BONDING CONNECTIONS
Overhead HV bonding	32mm ² stranded copper, green PVC (2mm radial thickness) insulated conductor
Overhead LV side and Neutral bond	32mm ² stranded copper, black PVC (2mm radial thickness) insulated conductor
Between LV Overhead line and cable	32mm ² stranded copper, Black PVC insulated
Underground substation between high voltage equipment	70mm ² stranded copper, PVC insulated conductor or insulated copper strip 3mmx25mm
Underground substation main LV neutral to earth electrode	70mm ² stranded copper, green/yellow PVC insulated conductor
Connection between sheath of SNE cable and Neutral of CNE cable	70mm ² stranded copper (2 x 35mm ²), green/yellow PVC insulated conductor for cable sizes above 70 mm ² for smaller size cable at least equivalent to the phase conductor with a minimum size of 16mm ²
Bonding and Earth connection to LV earth electrodes on underground cable networks	25mm ² , stranded copper, green/yellow PVC insulated conductor

Table 1 Earthing and bonding connections for different locations

Main bonding at customer's premises (See table below)

Connection location	Copper equivalent cross-sectional area
At customer's premises, connection between service neutral and company's earthing terminal	Green/yellow 16mm ² or half the size of the Company's neutral meter tail, whichever is the larger
At customer's premises connection between the company's earthing terminal and the earth bar of the consumer unit.	Green/yellow 16mm ² or half the size of the Company's neutral meter tail, whichever is the larger

Table 2 Main bonding at customer's premises

Appendix E Type and Size of Earth Connections to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	70 of 87

Main bonding equipotential connections between earthing terminal and all metallic structures (See table below).

Copper equivalent cross-sectional area of Supply neutral conductor	Minimum copper cross-sectional area of bonding conductor (or continuously rated equivalent)
35mm ² or less	10mm ²
over 35mm ² but not more than 50mm ²	16mm ²
over 50mm ² but not more than 95mm ²	25mm ²
over 95mm ² but not more than 150mm ²	35mm ²
over 150mm ²	70mm ²

Table 3 Size of Main Equipotential Bonding Copper Conductors

Notes:

- 1) Supplementary local bonding conductors and circuit protective conductors in the customer's installation should be of the size specified in the current edition of BS7671 (IEE Wiring Regulations)
- 2) Conductors on poles above 3m need not be insulated
- 3) Where copper theft has occurred on a site the above ground bonding should be replaced with aluminium conductor (see note 6), with suitable bi-metallic joints at the interface between the aluminium and the copper in the ground.
- 4) All earthing and bonding connections where buried direct in the ground shall be of copper conductor. Where copper to aluminium connections are made the aluminium conductor and joint shall be protected against corrosion by wrapping with grease impregnated tape.
- 5) To avoid possible dangerous ground potentials an earth electrode shall be installed so that its topmost point will be at least 600 mm below ground level. The insulation of the HV or LV earth-conductor shall be retained up to the earth electrode terminal.
- 6) The copper equivalent cross-sectional areas of Supply neutral conductors for typical DNO cables are as follows:
35 mm² Al CNE – 22 mm²
95 mm² Al CNE – 60 mm²
185 mm² Al CNE – 116 mm²
300 mm² Al CNE – 116 mm²
300 mm² Cu CNE – 150 mm²
4mmx25mm Cu tape – 6mmx25mm Al tape
4mmx40mm Cu tape – 6mmx40mm Al tape

Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	71 of 87

Contents

Contents	Page Number
1. Purpose	1
2. Scope	1
3. Policy	1
3.1 Assessment of Relevant Drivers	1
3.1.1 Requirements of the Electricity Act 1989	2
3.1.2 Requirements of the Electricity Safety, Quality and Continuity (ESQC) Regulation 2002	2
3.1.3 The Health and Safety at Work etc Act 1974	5
3.1.4 Electricity at Work Regulations	5
3.2 Key Requirements	5
3.2.1 New Networks	5
3.2.2 Existing Networks	6
3.3 Implementation	6
3.3.1 New Networks	6
3.3.2 Existing Networks	6
3.4 Provision of CE Earth Terminals	7
3.4.1 Provision of CE earth terminals on networks with electrically continuous sheaths or separate earth conductors	7
3.4.2 Provision of CE earth terminals on networks to which PME has been applied	8
3.4.3 Adequacy of Existing Bonding	8
3.4.4 Provision of an earth terminal after a cut-out change	8
3.4.5 Short-term safeguards for customers on substantial existing networks without adequate earth-return facilities	8
3.5 PME and Multiple-Earthing Approvals	9
3.6 Maximum Earth Values at Secondary Distribution Substations	9
3.7 Earthing Arrangements On New LV Networks	9
3.7.1 New Underground Networks	9
3.7.2 New Overhead Networks	10
3.8 Earthing Arrangements on Existing LV Networks	11
3.8.1 Underground Networks with Electrically Continuous Sheaths	11
3.8.2 Underground Networks with Electrically Discontinuous Sheaths	14
3.8.3 Overhead Networks with Aerial Earth Wires	14
3.8.4 Overhead Networks without Aerial Earth Wires and which have not been converted to PME or PNB	15
3.8.5 Overhead and Underground Networks which have been converted to PME	15
3.9 Earth Electrodes on PME Systems	15

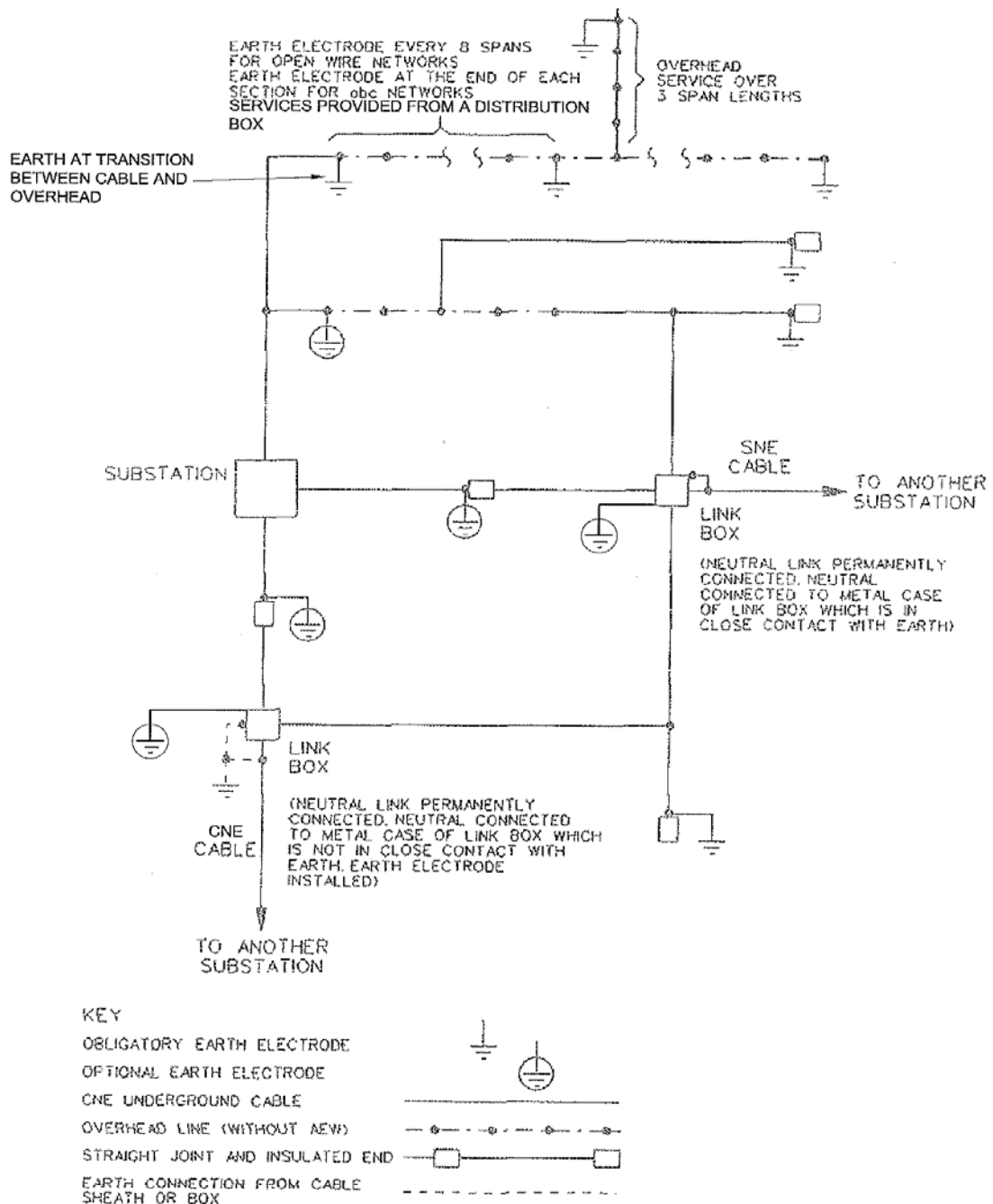
Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	72 of 87

3.10	Arrangements at Link Boxes and Pillars on PME Networks	16
3.11	Street Lighting and Other Street Furniture	17
3.11.1	The application of PME	17
3.11.2	Street Lighting Brackets Mounted on Poles or Buildings	18
3.11.3	Roadside and Other Housings Accessible to the Public	18
3.11.4	Existing Separate Street Lighting Mains with Inadequate Earth Return	18
3.11.5	Street Lighting Services from Electrical Networks with Inadequate Earth Return	18
3.11.6	Street Lighting Columns with Inaccessible Cable Shafts	19
3.12	Supplies to Customers Direct from Substations	19
3.12.1	Ground Mounted Substations	19
3.12.2	Pole Mounted Substations (PNB)	19
3.13	Customer's Earthed-Sheath-Return Wiring Installation or Customer's Distribution Installation Using Combined Neutral and Earth Conductors	20
3.14	Non-Standard Networks with Inadequate Neutrals	20
3.15.	Customers Installations Requiring Special Attention	20
3.15.1	Construction Sites	21
3.15.2	Farms, Milking Parlours, Pig Stys, etc	21
3.15.3	Swimming Pools	22
3.15.4	Sports Pavilions	22
3.15.5	110V Centre-Tapped Supplies via an Isolating Transformer	23
3.15.6	Quarries	23
3.15.7	Petrol Filling Areas	23
3.15.8	Fairgrounds and Showgrounds	23
3.15.9	Rail Service Areas	23
3.15.10	Caravans, Mobile Homes, Temporary Site offices, Boat Installations and Outside Broadcast Vehicles	23
3.15.11	High-Rise Buildings and Building Complexes	25
3.15.12	Supply terminating in a Separate Building	26
3.15.13	PME and Outside Water Taps	26
3.15.14	Lightning Protection Systems	27
3.15.15	Roadside and Other Housings Accessible to the Public	27
3.16	Under-Eaves Wiring Installations	28
3.16.1	New Under-Eaves Wiring Installations	28
3.16.2	Existing Under-Eaves Wiring Installations	28
3.17	Customers' Private Generator Installations	29
3.17.1	PME/PNB Installations	29
3.17.2	Non-PME Installations	29
3.18	Connections to Mobile Phone Base Stations associated with Transmission Towers or Primary Substations	29
3.19	Cathodic Protection Installations.	30
3.20.	Earthing at HV Distribution Substation	30
3.20.1	Earthing Requirements at new Ground Mounted Distribution Substations	30
3.20.1.1	Introduction	30
3.20.1.2	General Earthing Requirements	30

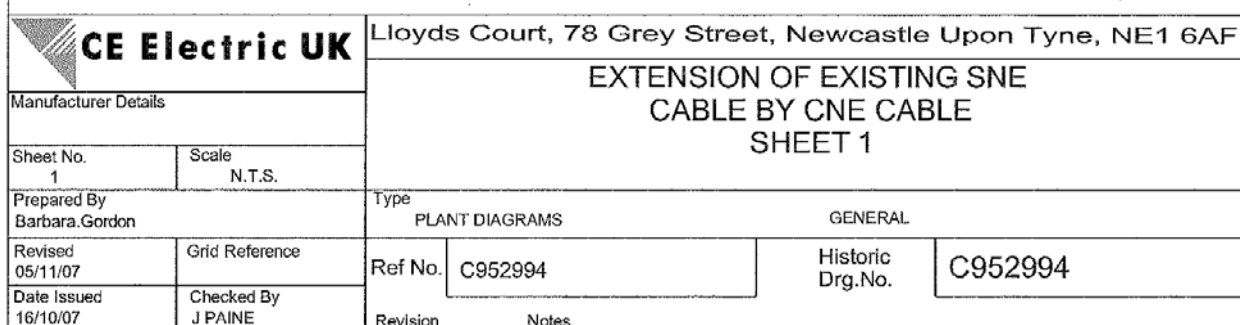
Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	73 of 87

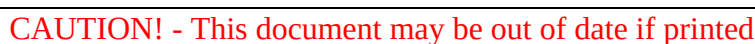
3.20.2	Bonding and Control of Potential	31
3.20.2.1	General	31
3.20.2.2	Bonding	32
3.20.2.3	Control of Potential	32
3.20.3	Distribution Substaion near or within a high Voltage Substaion or Tower	32
3.20.4	Customer Substations	32
3.20.5	Combined HV/LV Earths and the Global Earthing Concept	33
3.20.6	Construction Details	38
3.20.6.1	Materials	38
3.20.6.2	Installation of Electrodes	38
3.20.6.3	20/11/6.6kV Cables	39
3.20.7	Standard Arrangments	39
3.20.7.1	Standard Arrangement Drawings	40
3.20.7.2	Standard Arrangement Resistance Values	40
3.20.7.3	Separate LV Earth Arrangments	41
3.20.7.4	Standard Earthing Arrangemts for Brick or Similar Build Substations	42
3.20.7.5	Standard Earthing Arrangemts for Steel Framed or Kiosk Substations	43
3.20.8	Pole-Mounted Substations	48
4	References	50
4.1	Intrernal	50
4.2	External	50
5	Definitions	52
6	Authority for Issue	54
Appendix A	Customers' Installations	55
Appendix B	Procedure to be Applied where a Customer is Reported to have Inadequate Earths	58
Appendix C	Guidance on Minmising EMC Problems	65
Appendix D	Earth Electrode Testing	66
Appendix E	Type and Size of Earth Connections	68
Appendix F	Contents	70
Appendix G	General Drawings	73

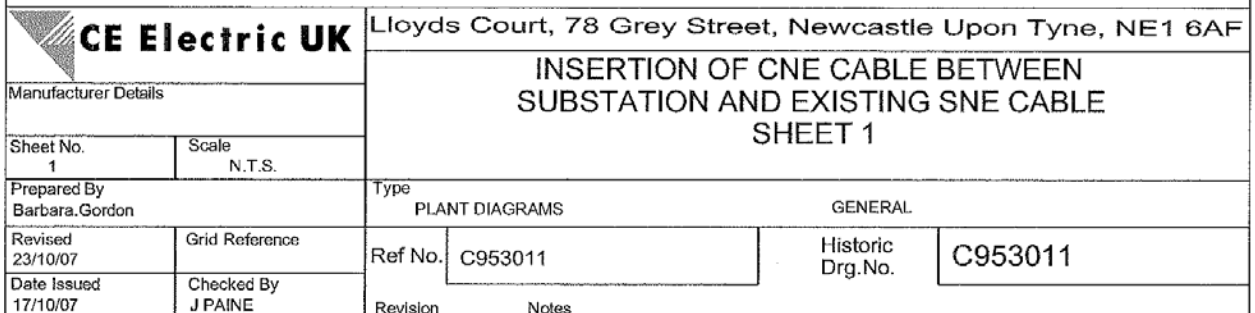
Appendix G General Drawings to Document		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 74 of 87



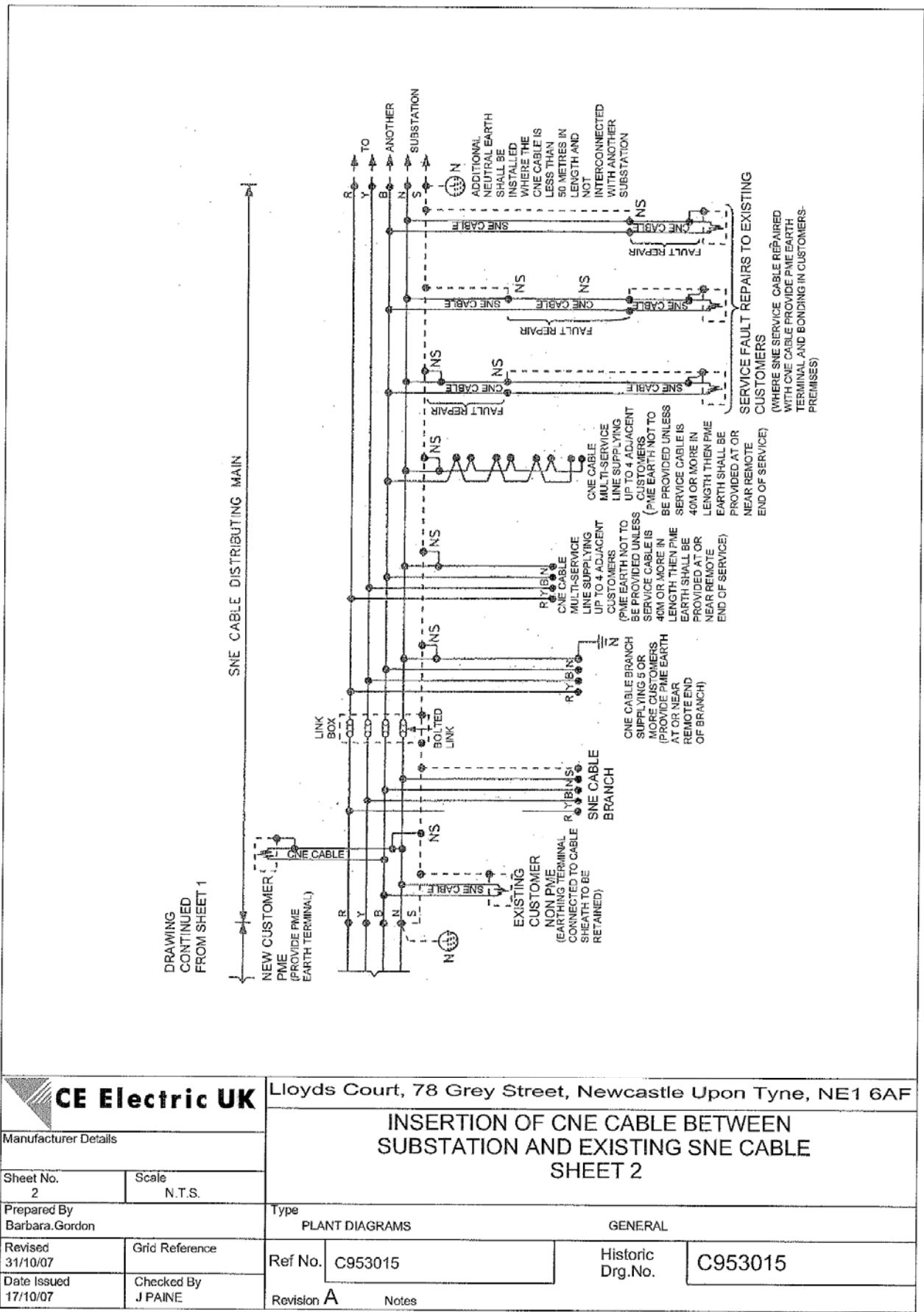
CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF		
Manufacturer Details		POSITIONS OF PROTECTIVE MULTIPLE EARTHING ELECTRODES ON NEW CNE UNDERGROUND CABLE NETWORKS AND NEW OVERHEAD LINES NETWORKS		
Sheet No. 1	Scale N.T.S.			
Prepared By Barbara Gordon		Type PLANT DIAGRAMS	GENERAL	
Revised 17/10/07	Grid Reference	Ref No. C952986	Historic Drg.No.	C952986
Date Issued 16/10/07	Checked By J PAINE	Revision	Notes	



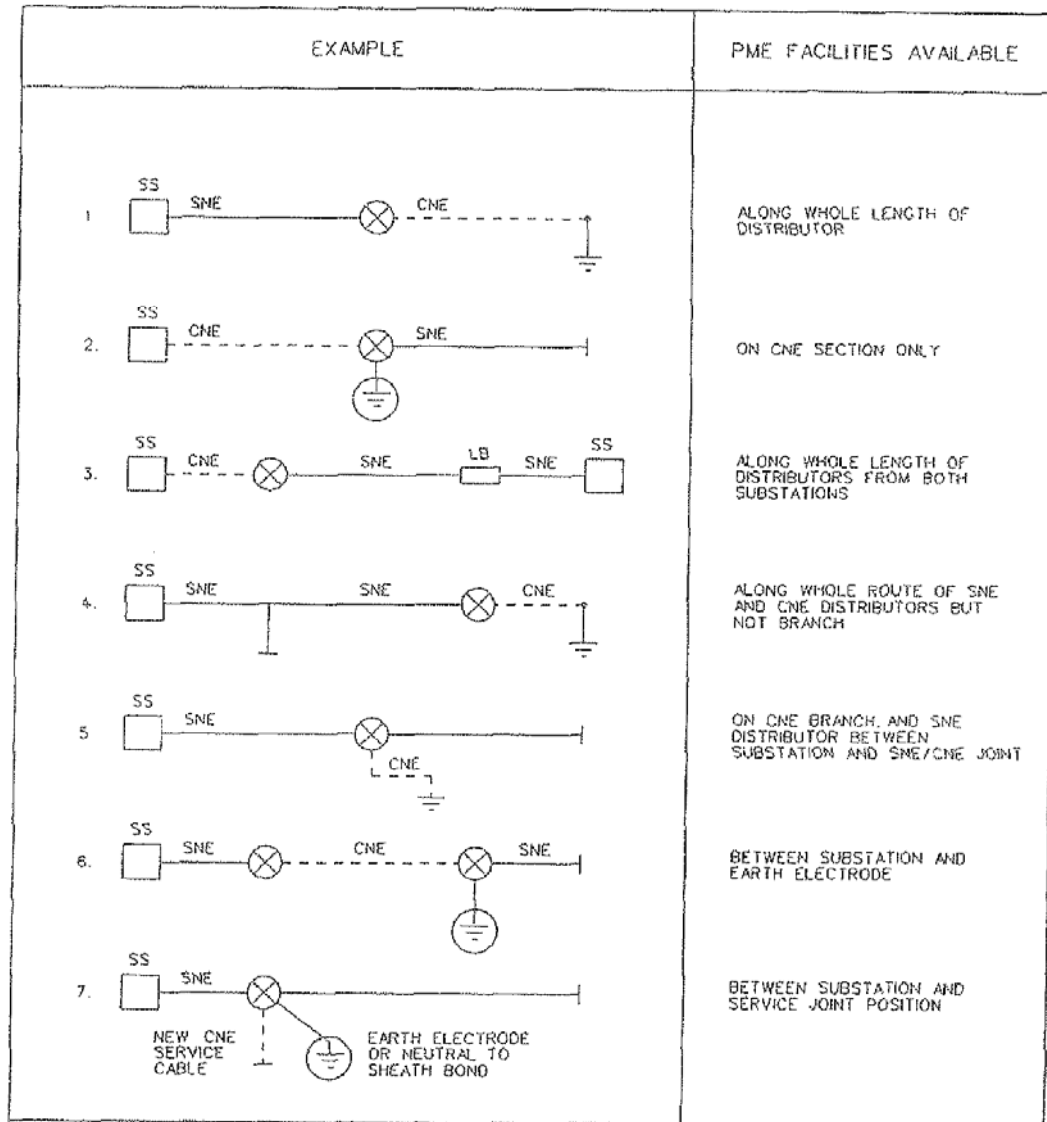




Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	78 of 87



Appendix G General Drawings to Document		IMP/010/011		
Version	1	Date of Issue	December 2007	Page 79 of 87



KEY

SS - SUBSTATION

CNE - COMBINED NEUTRAL AND EARTH CABLE

SNE - SEPERATE NEUTRAL AND EARTH CABLE

LB - LINK BOX (NEUTRAL SOLID)

(X) - BOND BETWEEN METALLIC SHEATH OF SNE CABLE AND NEUTRAL OF CNE CABLE



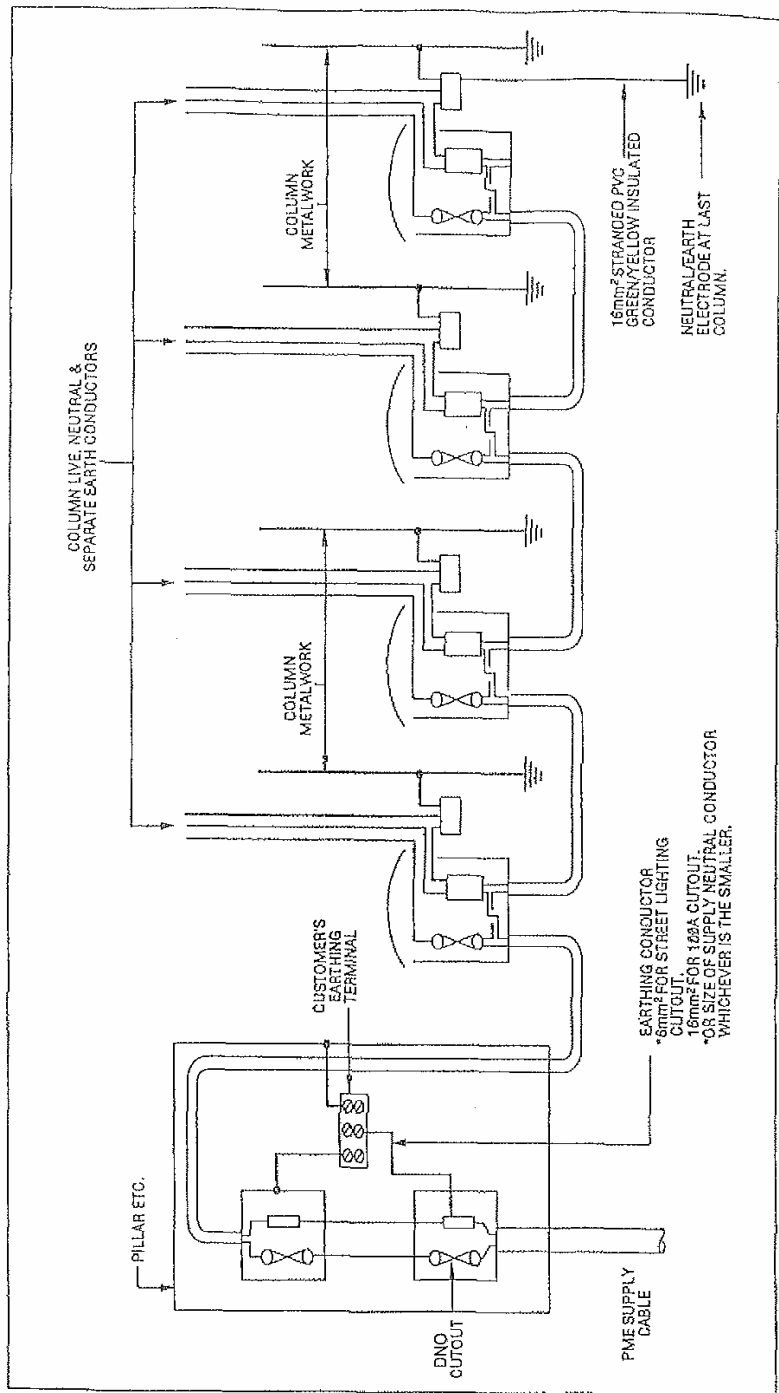
- OPTIONAL PME EARTH ELECTRODE



- COMPULSORY PME EARTH ELECTRODE

		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
		TYPICAL APPLICATIONS OF CNE CABLES	
Manufacturer Details			
Sheet No. 1	Scale N.T.S.		
Prepared By Barbara Gordon		Type PLANT DIAGRAMS	GENERAL
Revised 17/10/07	Grid Reference	Ref No. C953019	Historic Drg.No. C953019
Date Issued 17/10/07	Checked By J PAINE	Revision	Notes

Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	80 of 87



Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF

Manufacturer Details

STREET LIGHTING AUTHORITY
CNE DISTRIBUTOR FED FROM PME SERVICE

Sheet No. 1
Scale N.T.S.

Prepared By Barbara Gordon

Type PLANT DIAGRAMS GENERAL

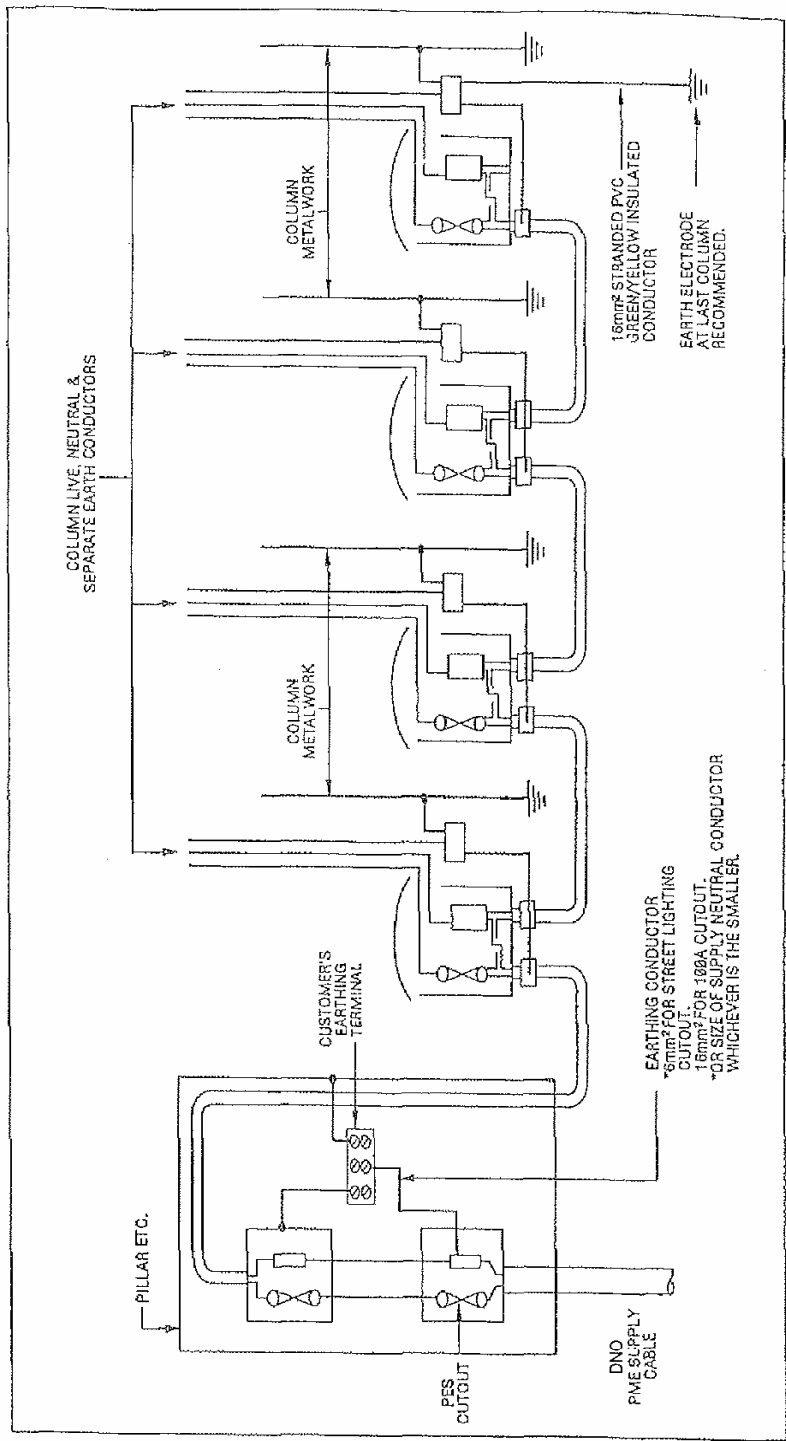
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Grid Reference

Ref No. C953029
Historic Drg.No. C953029

Date Issued 17/10/07
Checked By J PAINE

Revision Notes

Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	81 of 87



Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF

Manufacturer Details

STREET LIGHTIN AUTHORITY SNE
DISTRIBUTOR FED FROM PME SERVICE

Sheet No. 1
Scale N.T.S.

Prepared By
Barbara Gordon

Type
PLANT DIAGRAMS GENERAL

Revised 17/10/07
Grid Reference

Ref No. C953031
Historic Drg.No. C953031

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Revision Notes

Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	82 of 87

FIG A OVERHEAD SUPPLY

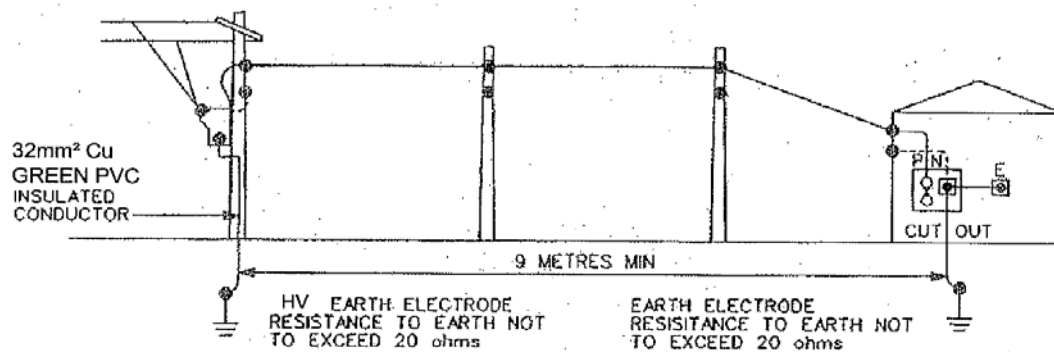


FIG B OVERHEAD SUPPLY WITH UNDERGROUND SERVICE CABLE

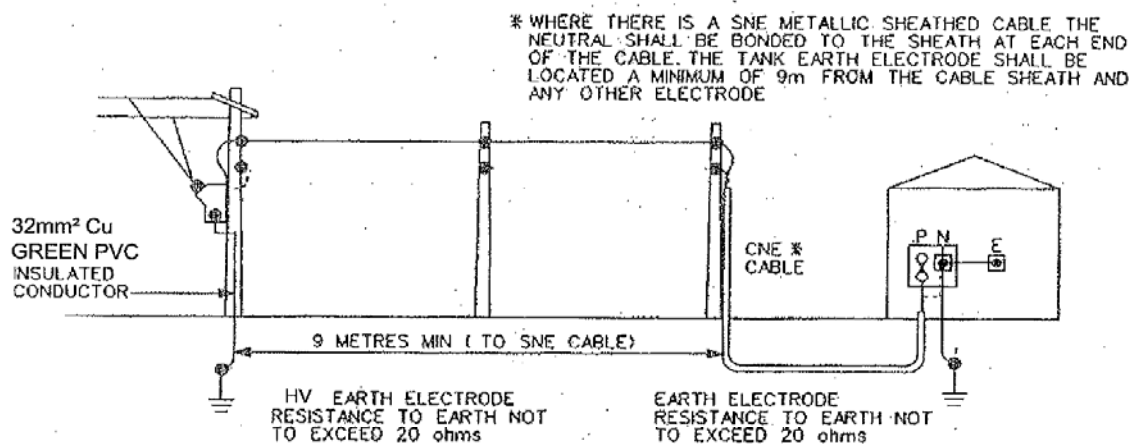
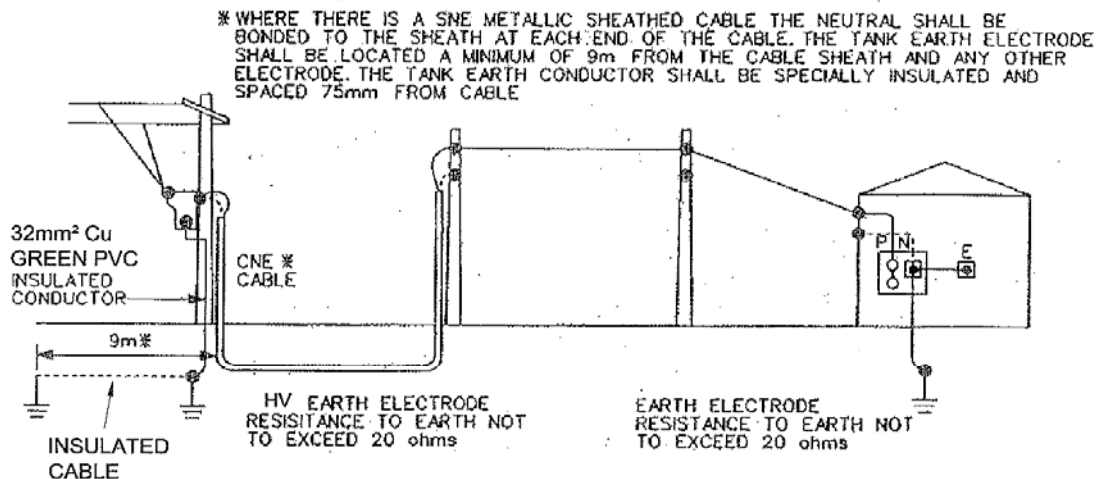
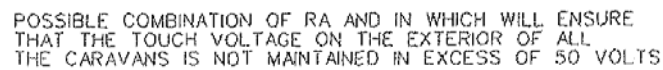


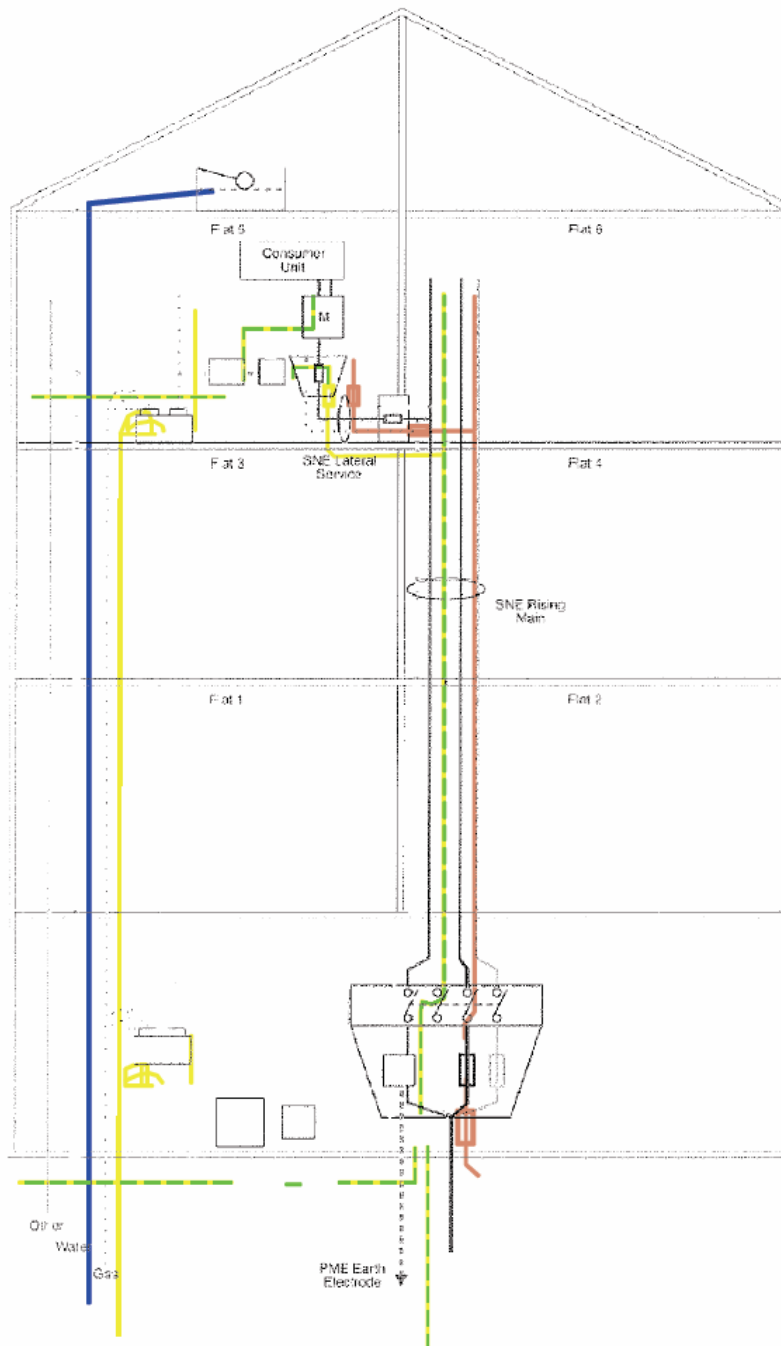
FIG C OVERHEAD SUPPLY WITH UNDERGROUND CABLE AT SUBSTATION



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		OVERHEAD SUPPLY TO ONE CONSUMER APPLICATION OF PROTECTIVE NEUTRAL BONDING P.N.B.			
Sheet No. 1	Scale N.T.S.				
Prepared By Barbara Gordon		Type PLANT DIAGRAMS	GENERAL		
Revised 17/10/07	Grid Reference	Ref No.	C953072	Historic Drg.No.	C953072
Date Issued 17/10/07	Checked By J PAINE	Revision	Notes		



Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	84 of 87



Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF

BONDING REQUIREMENT FOR SNE RISING MAIN

Manufacturer Details

Sheet No. 1 Scale N.T.S.

Prepared By
Barbara Gordon

Type
PLANT DIAGRAMS

GENERAL

Revised 17/10/07 Grid Reference

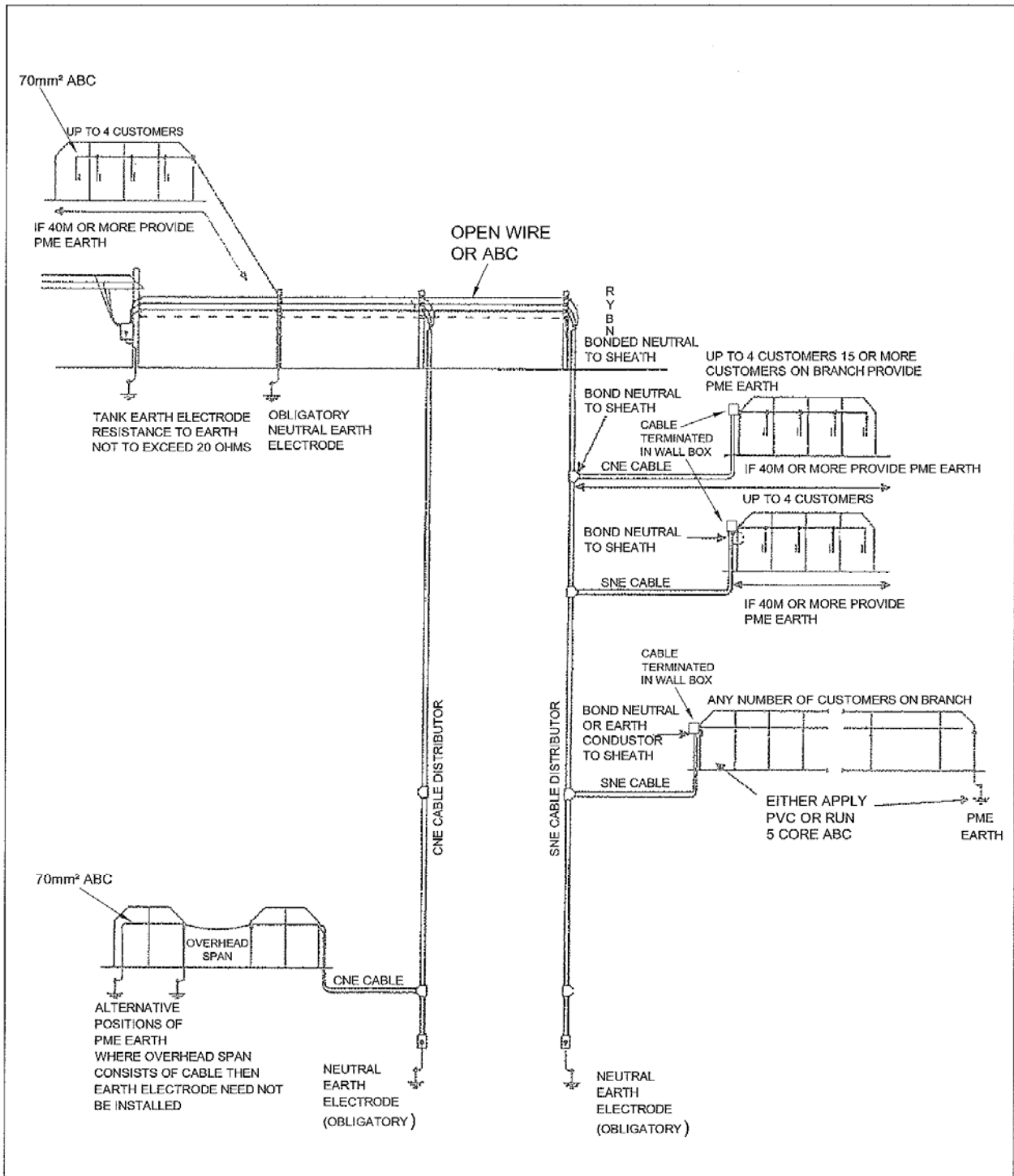
Ref No. C953070

Historic Drg.No. C953070

Date issued 17/10/07 Checked By J PAINE

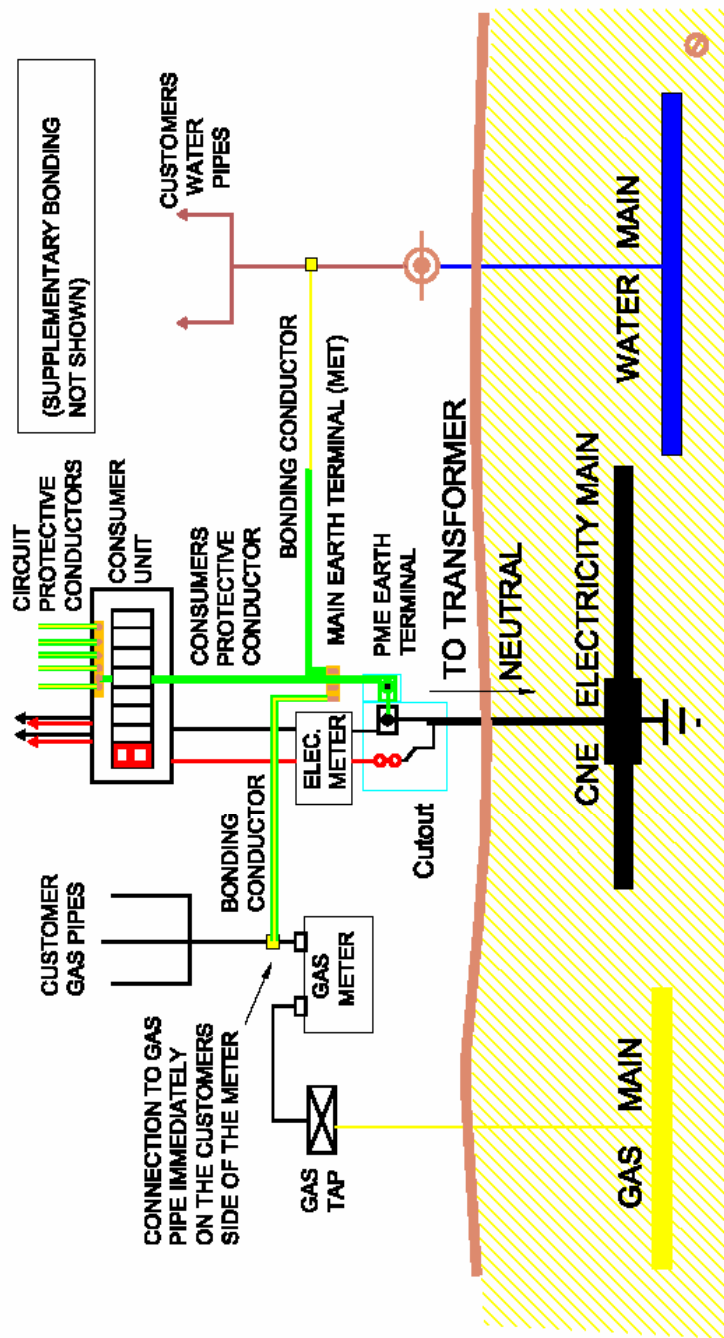
Revision Notes

Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	85 of 87



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		PME EARTHS ON UNDER-EAVES WIRING INSTALLATION			
Sheet No. 1	Scale N.T.S.	Type PLANT DIAGRAMS GENERAL			
Prepared By Barbara Gordon		Ref No. C953074 Historic Drg.No. C953074			
Revised 22/10/07	Grid Reference	Revision Notes			
Date Issued 17/10/07	Checked By J PAINE				

Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	86 of 87



Manufacturer Details

Sheet No. 1 Scale N.T.S.

Prepared By Barbara Gordon

Revised 17/10/07 Grid Reference

Date issued 17/10/07 Checked By J PAINE

Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF

EXAMPLE OF DOMESTIC MAIN BONDING

Type PLANT DIAGRAMS

CIRCUIT

Ref No. C953036

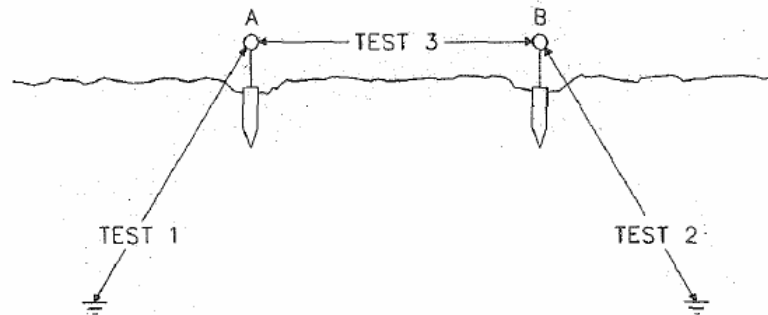
Historic Drg.No.

C953036

Revision Notes

Appendix G General Drawings to Document		IMP/010/011			
Version	1	Date of Issue	December 2007	Page	87 of 87

CHECKING SEPARATION OF TWO EARTH ELECTRODES



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
		EARTH ELECTRODE TESTING			
Manufacturer Details					
Sheet No. 1	Scale N.T.S.				
Prepared By Barbara Gordon		Type PLANT DIAGRAMS		GENERAL	
Revised 17/10/07	Grid Reference	Ref No.	C953068	Historic Drg.No.	C953068
Date Issued 17/10/07	Checked By J PAINE	Revision	Notes		